

LUPINE SYSTEMS

Study Sessions & Market Readings

V0 theory – complete (Parts I–IV) · plus annotated readings from Wise, Adyen & Remitly prospectuses

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PART I

Study Sessions

Complete V0-essential theory across Parts I–IV — the legacy problem, routing architecture, scoring engines, execution and cryptographic

evidence. Diagrams and comprehension checks preserved as recorded.

PART I — CHAPTERS 1-2

The Legacy Problem & How Cross-Border Money Moves

Completed prior to 8 Apr 2026

These two chapters lay the foundation for why Lupine must exist. Chapter 1 maps the deep structural limitations of the existing global payments framework. Chapter 2 exposes the true mechanics of cross-border flows.

CORE THESIS — CHAPTER 1

The global payments system was built for a world that didn't need real-time settlement, biological timing, corridor intelligence, or deterministic routing. It is fundamentally reactive, not predictive.

Research and medical ecosystems depend on precise timing, predictability, and traceability. Yet the underlying money-movement architecture was engineered for none of these. A "simple" cross-border payment actually traverses compliance checks, correspondent banks, treasury desks, and local clearing — all with different rules, operating hours, and liquidity constraints.

Routing is never precomputed or optimised. It's a chain of manual or static preferences set years ago. No global intelligence layer evaluates probability of delay, corridor reliability, predicted compliance flags, time-window compatibility, multi-hop failure likelihood, FX exposure across delays, or liquidity constraints during weekends and holidays.

Settlement window misalignment is a major cause of delays. RTGS systems across countries operate on different hours:

- AU RITS: 08:00–18:00
- EU TARGET2: 17:00–02:00 AEST
- US Fedwire: 21:00–05:30 AEST

There is no global window where all systems are open. A payment that misses a window gets held 24+ hours.

Regulatory friction is the #1 cause of unpredictable delays. Medical/research payments trigger red flags because they involve international transfers, restricted equipment, biological agents, and cross-border grants. Each country applies different AML/CTF rules, and no two banks interpret compliance the same way.

The book identifies **nine structural failure modes** in legacy systems: corridor volatility, routing ambiguity, manual review saturation, liquidity siloing, MT message desync, no predictive engine, evidence black holes, biological timing mismatch, and zero determinism.

CORE THESIS — CHAPTER 2

Cross-border money movement is a multi-layer consensus problem across conflicting networks — different hours, liquidity rules, compliance regimes, message formats, and failure surfaces. The system was never designed for determinism.

Chapter 2 exposes the **five-layer architecture** of global payments:

- **Layer 5 — Compliance:** AML/CTF, sanctions, KYB, travel rule
- **Layer 4 — Settlement:** RTGS, ACH, instant payment systems
- **Layer 3 — Correspondent network:** Nostro/vostro, treasury, liquidity corridors
- **Layer 2 — Messaging:** SWIFT MT, gpi, ISO 20022
- **Layer 1 — Funding:** Primary accounts, ledgers, balances

Payments move horizontally from sender to receiver, but the real work happens vertically across these layers.

Correspondent banking is the real backbone. No bank has accounts everywhere, so payments hop through 1–4+ correspondent banks. Each hop adds latency, compliance friction, and liquidity dependency. The path is determined by historical agreements and static rules — not computed intelligence.

Nostro/vostro: Nostro = "our money held with them." Vostro = "their money held with us." These prefunded accounts must absorb payment volume, FX volatility, and demand spikes. When liquidity runs out, payments queue.

Compliance at every hop: Every correspondent performs its own sanctions screening, AML scoring, and purpose validation. A payment clean at origin can be flagged by any intermediate bank. This is the #1 source of multi-day delays.

Real corridor latency (AU → SG → DE): Fastest: 2.5 hours. Median: 28–36 hours. P95: 3–5 days. Worst case: 7–11 days.

COMPREHENSION CHECK

Chris articulated the core Lupine value proposition:

CHRIS'S ANSWER

Legacy systems send payments and resolve failures reactively, whereas Aiva evaluates routes, predicts risks before they occur, and selects the optimal available route.

This correctly captures the fundamental distinction. Legacy = reactive. Lupine = predictive.

V0 CONNECTION

V0's route scoring (Aiva-lite) is the first working answer to failure modes 1, 2, 4, and 6. The evidence log (Clocked-lite) addresses failure mode 7. The V0 demo scenario (AUD 500K → SGD, high urgency) directly models the AU → SG corridor.

PART I — CHAPTER 3

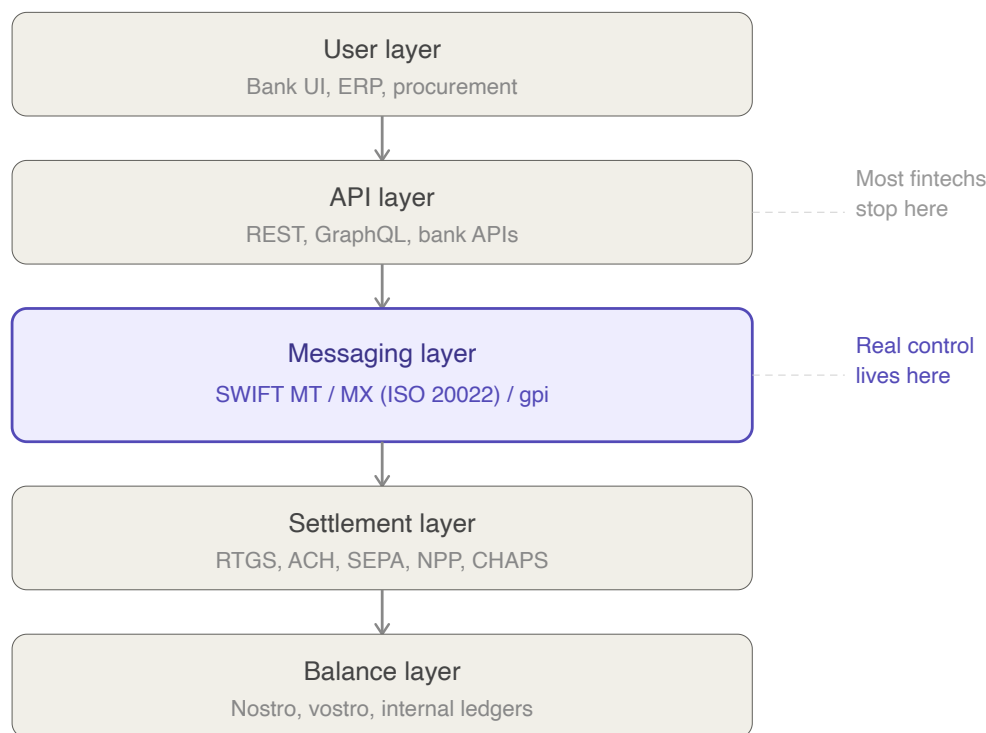
ISO 20022, SWIFT gpi & the Global Messaging Infrastructure

8 Apr 2026

The book calls this one of the most technical chapters in the entire Payments section. The core thesis: **messaging = meaning**. The entire global payments system is held together by messages, and those messages are the real control layer.

KEY INSIGHT

Most fintechs and banks only touch the API layer. But the messaging layer is where the real power sits — because SWIFT messages aren't just data packets, they're legal and operational instructions that tell every system in the chain what to do.

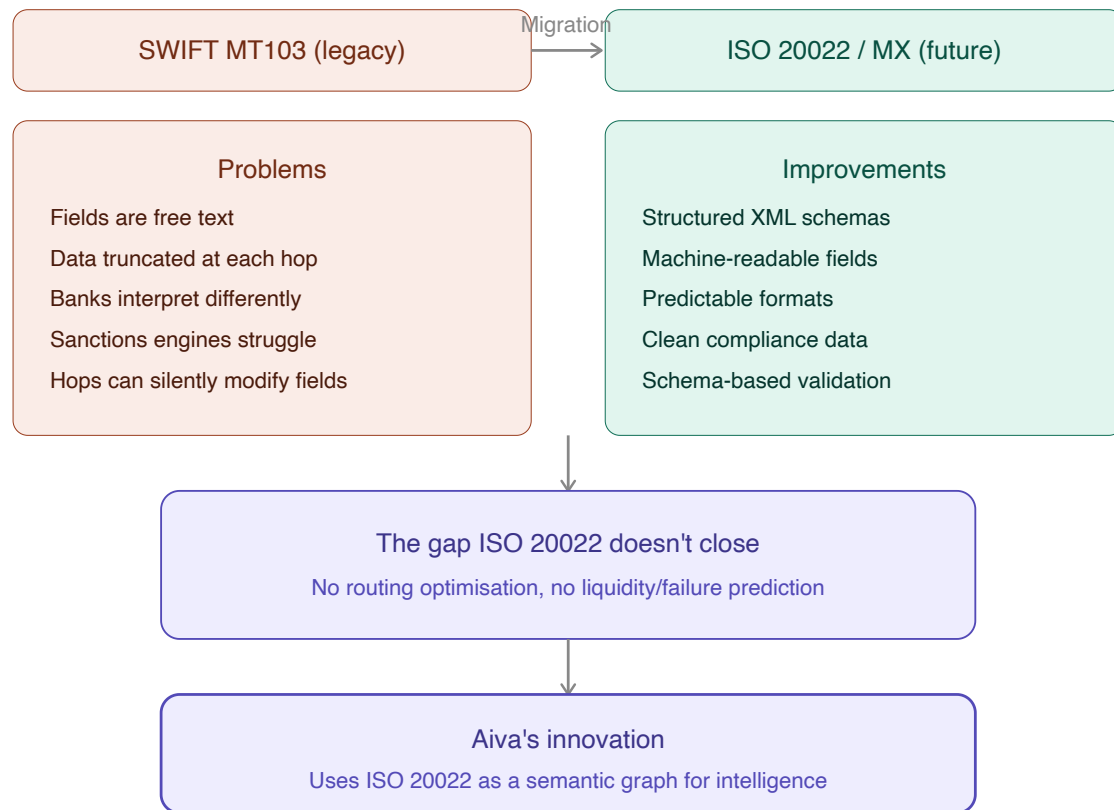


Messages are legal, regulatory, and operational instructions
When the messaging layer is weak, the entire system is unstable

The chapter walks through **three generations** of the messaging layer:

- **SWIFT MT** (legacy, still running the world) — free-text, fragile, ambiguous. Fields get truncated at each hop, banks interpret differently, sanctions engines struggle with unstructured data.

- **ISO 20022 / MX** (the structured future) — XML-based, machine-readable, compliance-ready. Strict schemas, predictable formats, schema-based validation.
- **SWIFT gpi** (tracking overlay) — adds end-to-end tracking, hop-level timestamps, fee transparency. But does NOT solve liquidity shortages, compliance blocks, routing intelligence, or predictive failure.



SWIFT gpi — visibility, not intelligence. gpi gives you timestamps at each hop. You can see that your payment sat at the Singapore correspondent for 90 minutes waiting for liquidity. But gpi can't predict that was going to happen, can't reroute around it, and can't score whether a different corridor would have been faster. It's a rear-view mirror. Aiva is the windshield.

The Lupine innovation: Banks use ISO 20022 for migration compliance and data quality. Lupine uses it for corridor scoring, routing optimisation, predictive failure simulation, FX exposure modelling, compliance pre-checks, and evidence chain construction. A payment becomes six things simultaneously in Aiva's model: a constraint set, a risk vector, a compliance object, a corridor graph node, a time-window object, and a liquidity demand event.

Evidence capsules (Cloked layer): Cloked extracts structured ISO 20022 fields into cryptographic evidence capsules — hashed party IDs, anchored timestamps, compliance scores, jurisdiction chains. This is only possible because ISO data is structured and schema-validated. You can't hash the ambiguity of MT103.

COMPREHENSION CHECK 1

The book says "cross-border payments do not run on SWIFT." What does the book actually mean? What does SWIFT really do versus the systems that move the money?

CHRIS'S ANSWER

Cross-border payments run on messages across different institutions. SWIFT encapsulates the value information between the interacting financial institutions and makes sure that the information is visible at each stage.

Close — but a key sharpening: SWIFT itself is just the messaging network — the postal service, not the money. The actual money moves through RTGS systems (RITS, TARGET2, Fedwire), nostro/vostro balances, FX desks, and clearing houses. SWIFT carries the instructions that tell all those systems what to do. It doesn't move a single dollar.

COMPREHENSION CHECK 2

Why does the book say gpi is "visibility, not intelligence"?

CHRIS'S ANSWER

gpi is visibility and not intelligence because it does not solve liquidity problems, compliance blocks, does not have any kind of intelligence, predictive nature, or failure scoring.

Nailed it. gpi tracks problems — it does not prevent them. That distinction is the entire Lupine thesis.

COMPREHENSION CHECK 3

Why are evidence capsules only possible with ISO 20022 and not with MT103?

CHRIS'S ANSWER

MT103 format is unstructured and cluttered, whereas ISO 20022 data is rich and has comprehension that can be computed and analysed.

Correct. MT103 fields are free text — addresses, names, and purpose codes crammed together with no programmatic separation. ISO 20022 gives discrete, labelled, schema-validated fields. You can hash individual structured fields and build verifiable evidence. You can't hash ambiguity.

V0 CONNECTION

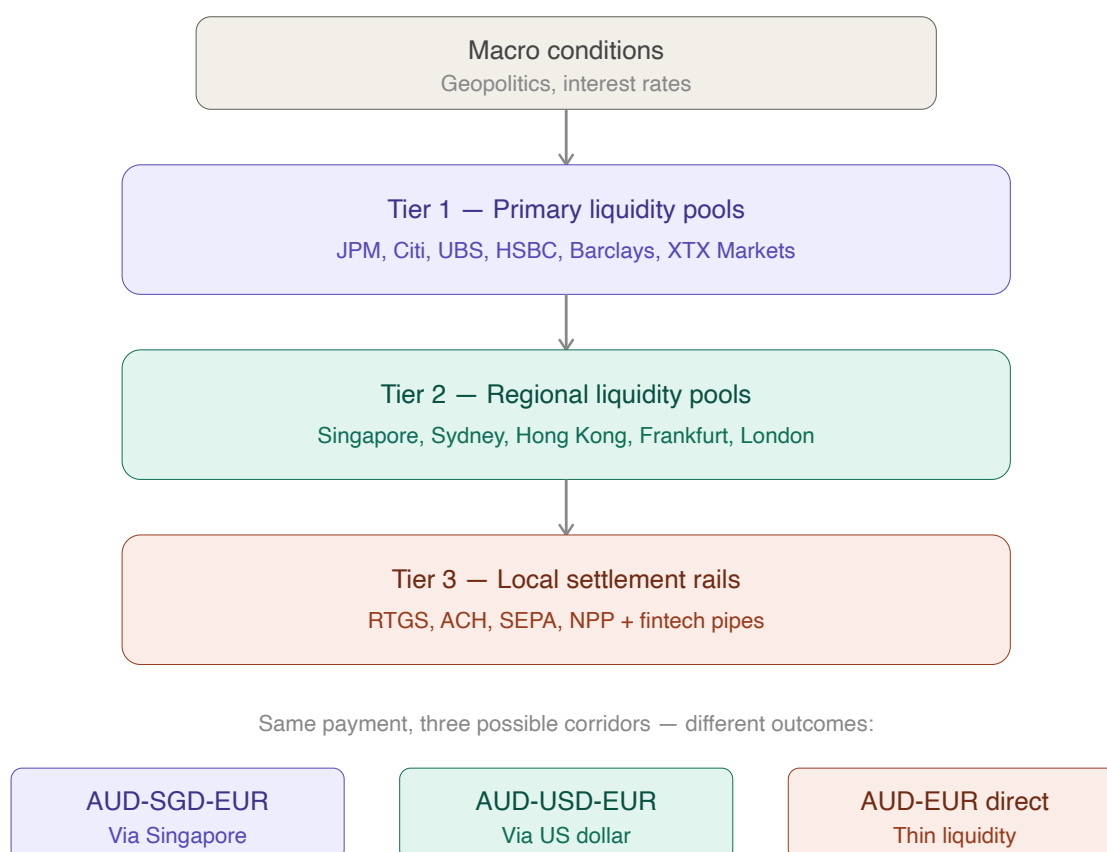
Aiva-lite's scoring model is the simplified version of "ISO 20022 as semantic graph." In V0, route options carry structured metadata (speed, cost, reliability) that Aiva-lite reasons over. Cloked-lite's evidence log is the first implementation of the evidence capsule concept.

PART I — CHAPTER 4

FX Intelligence & Corridor Engineering

9 Apr 2026

This chapter directly feeds how Aiva-lite constructs and scores routes in V0. The big idea: cross-border payments don't move randomly. They move through **corridors** — geopolitical, regulatory, liquidity-dependent pathways between currencies. The same payment can take completely different corridors with wildly different outcomes.



The real FX rate. Everyone thinks the rate is just "AUD/EUR = 0.59." The book says that's wrong. The actual rate is built from 8 stacked components:

$$\text{Effective rate} = S_0 + dS + F + V + L + C + D + R$$

S₀	Interbank spot rate — what banks trade at scale
dS	Spread — margin from liquidity providers
F	Forward points — interest rate differentials
V	Volatility premium — risk during movement
L	Liquidity premium — wider spread when thin
C	Corridor cost — costs unique to this path
D	Execution delay premium — price impact from latency
R	Compliance risk buffer — added for high-risk corridors

Hidde

Banks rarely expose V, L, C, D, R — fintechs expose some

Lupine exposes all 8, at the intelligence layer

Key: delay (D) increases volatility (V) which widens spread (dS)

The latency–FX exposure correlation is the chapter's most powerful insight:

$$\text{FX Exposure} = \sigma \times \sqrt{t} \quad (\text{volatility} \times \text{square root of settlement time in hours})$$

The square root means exposure grows non-linearly with time — the first few hours of delay hurt more than the next few. So Aiva isn't just picking the fastest route or the cheapest route. It's computing which route gives the **lowest total exposure** — the combination of time and volatility that produces the least price damage. A fast route through a volatile corridor could cost more than a slightly slower route through a stable one.

Aiva's corridor scoring model evaluates each corridor across six dimensions:

$$\text{Score} = w_1 \times \text{Liquidity} + w_2 \times \text{Latency} + w_3 \times \text{ComplianceRisk} + w_4 \times \text{Volatility} + w_5 \times \text{FXCost} + w_6 \times \text{Reliability}$$

Aiva evaluates 1-hop, multi-hop, synthetic, triangulated, and blended corridors, then selects the dominant one via Pareto frontier analysis. V0 simplifies this to three dimensions: speed, cost, reliability.

Banks vs fintechs vs Lupine: Banks route based on treasury cost optimisation — they use pre-funded nostro balances and existing correspondent agreements. They optimise for their cost, not your outcome. Fintechs avoid correspondents entirely (netting, internal ledgers, local rails). Lupine evaluates ALL available paths and optimises for the payment's outcome.

Seven corridor failure modes: liquidity evaporation, regulatory shock, market volatility spikes, window misalignment, correspondent collapse, treasury freeze, geopolitical fracture. Aiva predicts these by analysing volatility, liquidity depth, historical patterns, and compliance triggers — rerouting before collapse occurs.

COMPREHENSION CHECK 1

FX Exposure = $\sigma \times \sqrt{t}$. In your own words, what does this mean for how Aiva should pick routes? Why isn't "fastest route" always the right answer?

CHRIS'S ANSWER

The FX exposure means the best cost available at the best settlement time. When it comes to the fastest route, the fastest route might have more spread, which might increase exposure.

Right direction. Sharpening: σ is volatility, t is time. A fast route through a volatile corridor could actually cost more than a slightly slower route through a stable one. Aiva computes the lowest total exposure — the combination of time and volatility that produces the least price damage, not just speed or cost alone.

COMPREHENSION CHECK 2

Your V0 has Fast Route, Cheap Route, Balanced Route. Based on the 8 FX rate components — what's actually different between them?

CHRIS'S ANSWER

The main difference between the three routes is based on what the overall FX rate is built on given the corridor. Each will have different tradeoffs — some might have quick settlement time but volatility is more, in some cases the settlement time is less however the interest rate difference is huge.

Correct — each route occupies a different position on the Pareto frontier. Fast Route: deep Tier 1 pool, higher spread and corridor cost. Cheap Route: less liquid regional pool, lower spread but higher liquidity premium, delay premium, and volatility exposure. Balanced Route: minimises total exposure across all 8 components.

COMPREHENSION CHECK 3

Banks route based on "treasury cost optimisation" while Lupine routes based on "outcome optimisation." What's the actual difference?

CHRIS'S ANSWER

Banks have pre-existing relationships with correspondents based on what they are charging each other, however the outcome optimisation is based on selecting all routes available including the correspondents.

Exactly. Banks are locked into existing correspondent relationships and pre-funded nostro accounts. They ask "what's cheapest for us?" Aiva has no such bias — it evaluates every corridor and picks the one that produces the best outcome for the payment itself.

V0 CONNECTION — DIRECT BLUEPRINT FOR AIVA-LITE

The 6-dimension scoring model simplifies to V0's speed/cost/reliability tradeoff. The three V0 routes represent different positions on the Pareto frontier. The 8 FX components explain WHY routes cost differently. The latency-exposure formula explains WHY faster isn't always better. This is the theoretical backbone of Aiva-lite.

PART II — CHAPTER 1

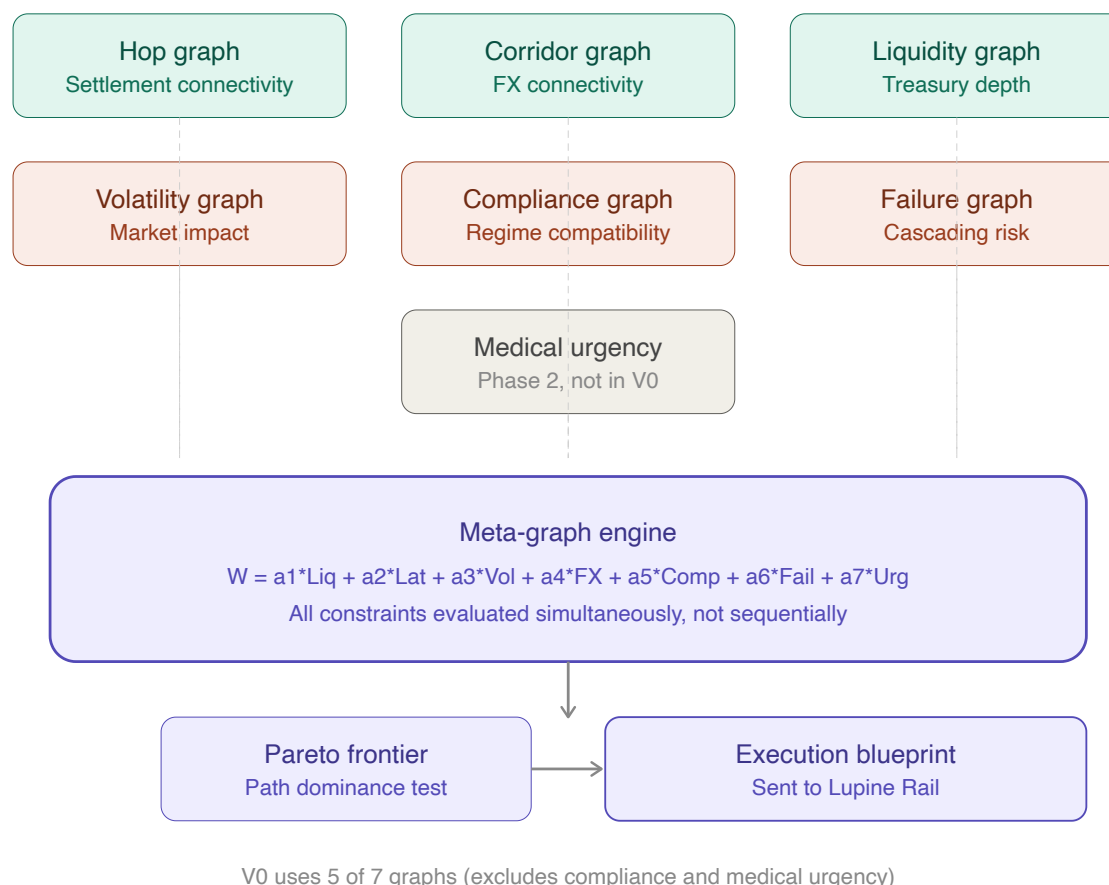
Deterministic Routing Architecture — Aiva's Core Brain

13 Apr 2026

This chapter is the architectural heart of Aiva. In Chapter 4 you learned *what* corridors are and *how* they're scored. This chapter explains *how Aiva's brain actually works* — the multi-graph intelligence system that makes routing decisions.

KEY INSIGHT

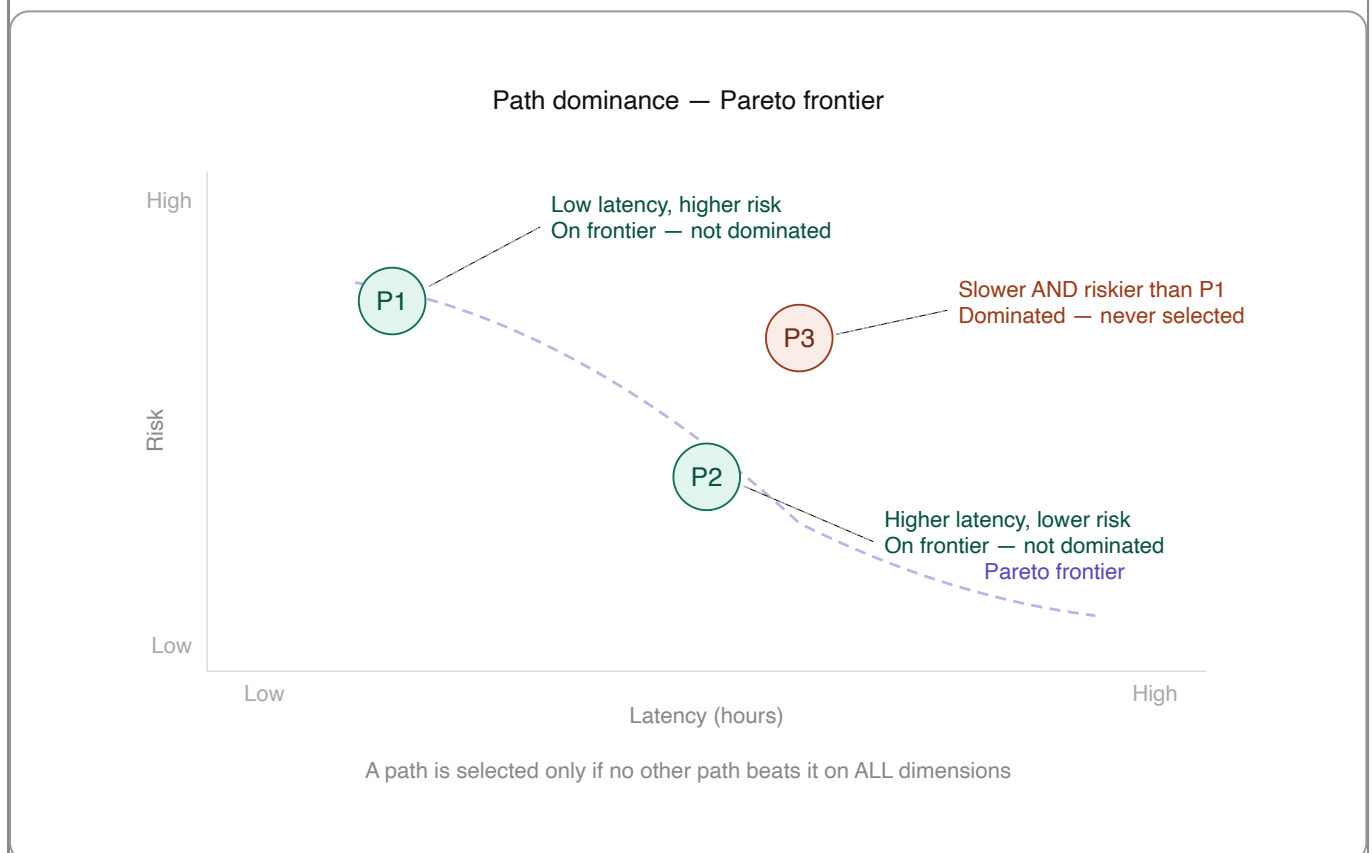
Aiva doesn't look at payments through one lens. It looks through seven simultaneously. Each one is a graph — a network of nodes and edges — and Aiva merges them all to compute the single best path.



What each graph does:

- **Hop graph** — maps which banks can reach which banks. The physical connectivity and settlement pathways. Edge weights: latency + reliability.
- **Corridor graph** — maps FX routes (AUD→SGD→EUR vs AUD→USD→EUR). Edge weights: liquidity depth, volatility penalty, spread. This is where Chapter 4's scoring lives.
- **Liquidity graph** — tracks treasury depth at each tier. Anticipates liquidity droughts and pool exhaustion before the payment hits.
- **Volatility graph** — tracks price risk propagation. Each edge carries σ , feeding into FX Exposure = $\sigma \times \sqrt{t}$.
- **Compliance graph** — maps which regulatory regimes are compatible. Predicts where payments will get flagged before they're sent.
- **Failure graph** — assigns failure probabilities to corridors. Predicts cascading failures using Bayesian models.
- **Medical urgency** — Phase 2 only. Ensures biological payloads avoid time-violating corridors.

The meta-graph merge is the critical concept. Legacy systems evaluate constraints sequentially: check compliance first, then routing, then liquidity. If compliance says no, stop. Aiva evaluates them simultaneously — computing a composite weight across all graphs for every possible path. This means Aiva can find paths that sequential systems would never discover, like a corridor that costs slightly more but avoids a compliance trigger entirely.



Path dominance: P1 is fast but carries more risk. P2 is slower but safer. Both are valid — they sit on the Pareto frontier because no other path beats them on every dimension. P3 is slower than P1 AND riskier than P2 — so it's dominated. Aiva would never pick it.

In V0 terms: Fast Route = P1, Balanced Route = P2, the route that doesn't get picked = P3. When the user sets urgency to "high," Aiva-lite weights toward P1. Normal urgency weights toward P2.

The execution blueprint: After all this computation, Aiva outputs a deterministic execution blueprint to Rail containing: optimal corridor path, hop-level timing model, liquidity requirements, compliance probability matrix, FX exposure curve, fallback tree, and evidence anchor map for Cloked. Rail executes it exactly as specified.

"Deterministic" means: given the same inputs, you get the same output every time. No randomness, no human judgment calls. Legacy systems are non-deterministic — two identical payments can take different paths. Lupine makes payments deterministic. This is possibly the most powerful single word for your Singapore pitch: "We make cross-border payments deterministic."

COMPREHENSION CHECK 1

Legacy systems evaluate constraints sequentially, Aiva evaluates them simultaneously. Why does this matter?

CHRIS'S ANSWER

In sequential evaluation, it is time consuming, and by removing this sequential evaluation, we can choose constraints based on faster outputs.

Speed matters, but the real problem is worse decisions. If compliance flags a payment first, it stops — it never reaches the routing engine, which might have found a corridor where that flag wouldn't have triggered. Sequential means each system decides in isolation, blind to the others. Aiva can find paths that sequential systems would never discover.

COMPREHENSION CHECK 2

What does it mean for a route to be "dominated," and why would Aiva never select it?

CHRIS'S ANSWER

P3 is dominated because this path is beaten both by P1 and P2. A route to be dominated is something that every other route surpasses.

Close — a route is dominated when at least one other route beats it on every dimension. P1 is faster AND lower risk than P3, so P3 can never be the right answer regardless of what you're optimising for. Routes on the Pareto frontier require tradeoffs — you give up something to gain something else.

COMPREHENSION CHECK 3

What does "deterministic" mean in this context, and why does the book keep using that word?

CHRIS'S ANSWER

Deterministic means simultaneously evaluating all the parameters for a desired output, which may be fast, reliable or safe, instead of sequential preference.

You're connecting it to simultaneous evaluation, which is related — but deterministic has a more specific meaning: given the same inputs, you always get the same output. No randomness, no human judgment in the middle. Legacy systems are non-deterministic — two identical payments can take different paths. Aiva's output is computed, not guessed. The outcome is predictable before the payment moves. "We make cross-border payments deterministic" — that's a sentence that would make any payments person lean forward.

V0 CONNECTION

V0 simplifies to 5 of the 7 graphs. Aiva-lite's speed/cost/reliability scoring is the simplified meta-graph merge. The three V0 routes are positions on the Pareto frontier. The "route selected because..." explanation is the V0 version of the deterministic execution blueprint. V0 proves the core thesis: computed routing, not reactive routing.

PART II — CHAPTER 2

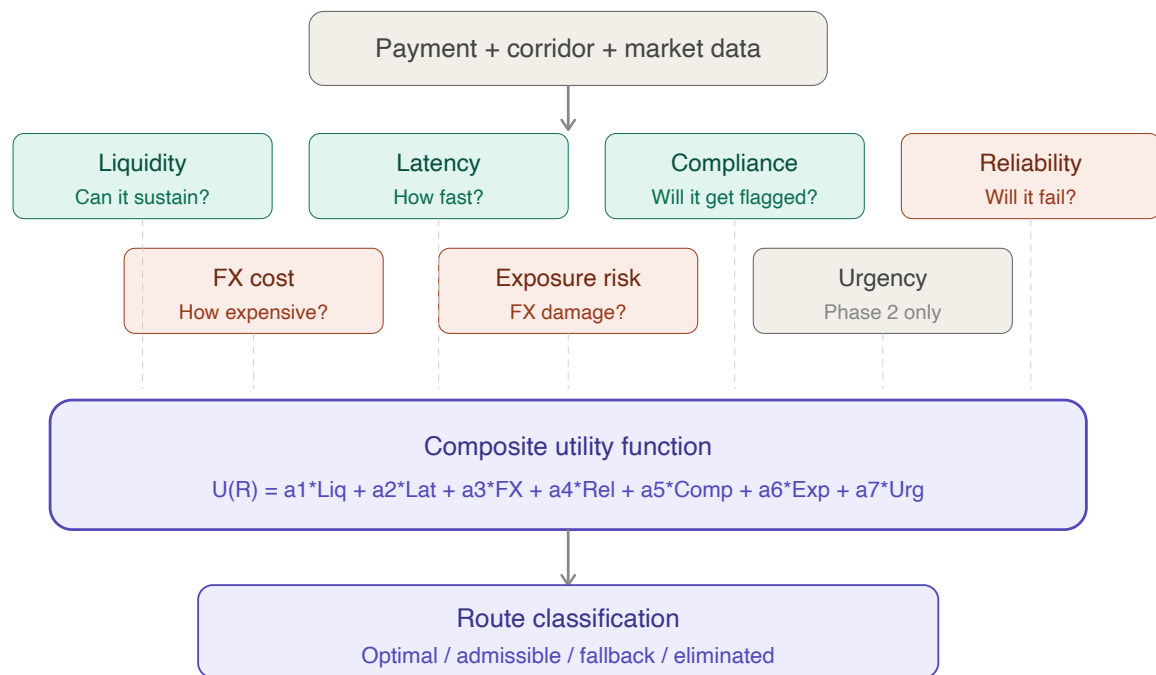
Scoring Engines — FX, Risk, Reliability, Compliance, Exposure & Urgency

14 Apr 2026

If P2-Ch1 was Aiva's brain architecture, this chapter is the reasoning machinery — how each of those seven graphs actually computes its score. Every route Aiva evaluates gets a **score vector** — not one number, but seven separate scores that together describe the route's characteristics.

KEY INSIGHT

The same scoring engine, evaluating the same corridors, can produce different "best" routes depending on the weights. When urgency is high, speed and exposure get heavy weights. When normal, cost and reliability dominate. The engine doesn't change. The weights do.



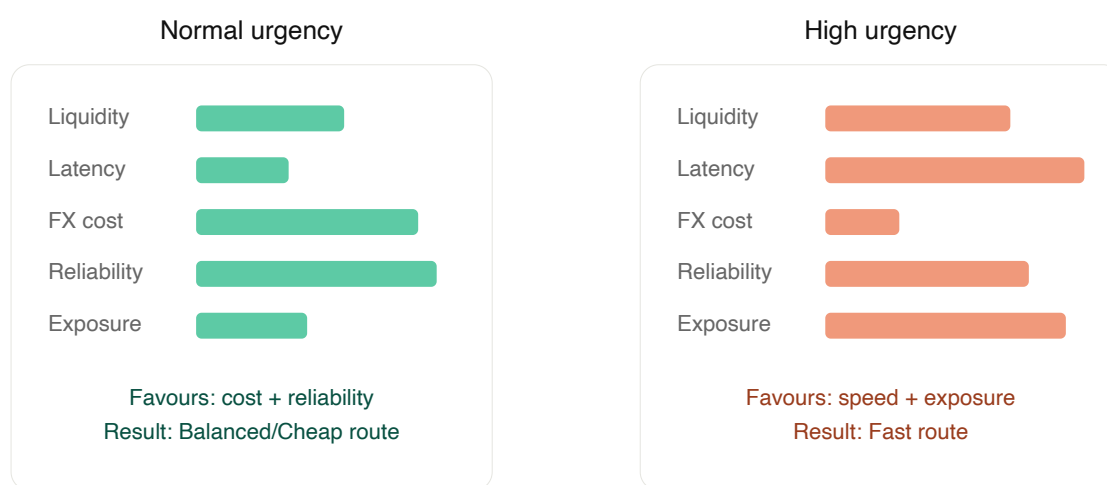
Weights shift based on urgency — same engine, different winner

The four V0-relevant scoring models:

- **Liquidity score (gatekeeper)** — $\text{LiquidityScore} = \beta_1 * \text{Depth} + \beta_2 * \text{Stability} + \beta_3 * \text{WindowFit} + \beta_4 * \text{VolatilityInverse}$. A corridor with no liquidity can't execute — eliminated before other scores matter. You can't score a bridge that's missing.
- **Latency score** — $\text{LatencyScore} = 1 / \text{ExpectedLatency}$. Predicted from RTGS windows, queue depth, correspondent behaviour, historical hop latency. Faster = higher score.
- **Reliability score** — $\text{ReliabilityScore} = 1 - P(\text{failure})$. Bayesian model: $P(\text{failure} \mid \text{corridor}, \text{time}, \text{liquidity}, \text{volatility})$. Banks never predict corridor failure before routing. Aiva does.
- **FX cost + Exposure risk (the pair)** — $\text{FXCost} = S_0 + \Delta S + C + D + V + L$. $\text{ExposureRisk} = \sigma \times \sqrt{t}$. These work together because delay increases exposure which increases cost. A cheap but slow route may score poorly overall.

The composite utility function assembles all scores with adjustable weights. High urgency: speed and exposure weighted up. Normal urgency: cost and reliability weighted up. Critically, **weights shift — they never zero out**. Even under high urgency, a route with 40% failure probability would still be rejected because reliability drags the total utility down.

Routes are classified into four tiers: **optimal** (highest U, selected), **admissible** (Pareto frontier, valid alternatives), **fallback** (used if optimal fails), **eliminated** (liquidity-starved or too risky, never selected).



Same engine, same corridors — different weights produce different optimal routes
This is exactly how VO's urgency parameter works

COMPREHENSION CHECK 1

The liquidity score acts as a "gatekeeper." Why does this make sense? What would go wrong if liquidity was treated the same as every other dimension?

CHRIS'S ANSWER

If liquidity was treated the same as every dimension there would be very less constraint to evaluate which corridor would be selected.

Right direction, but the real reason is more fundamental: a corridor with no liquidity literally cannot execute the payment. It doesn't matter if it has perfect latency, lowest FX cost, and clean compliance — if there's no money in the nostro account, the payment physically can't move. It's like scoring a bridge for speed and safety when the bridge is missing. Aiva eliminates impossible paths first to avoid computing scores for routes that can never work.

COMPREHENSION CHECK 2

The user inputs "high urgency" and V0 selects the Fast Route. What's actually happening under the hood? What changes and what stays the same?

CHRIS'S ANSWER

Speed and exposure is considered and cost and reliability is not considered.

Half right — speed and exposure do get weighted up. But cost and reliability are NOT ignored — they're still in the equation, just with lower weights. If a fast route had a 40% failure probability, Aiva wouldn't pick it even under high urgency because reliability would drag the total utility down. Weights shift, they never zero out. The pitch version: "We don't just pick the fastest route under pressure — we pick the best reliability-adjusted outcome under time pressure."

V0 CONNECTION

V0's three scores are the simplified composite. The urgency parameter is weight shifting. "High urgency" increases speed/exposure weights, "normal" increases cost/reliability. The three routes map to optimal/admissible. Eliminated routes don't appear. You can now explain exactly what's happening when the system says "Balanced Route selected because it gives the best reliability-adjusted outcome under high urgency."

PART II — CHAPTER 4

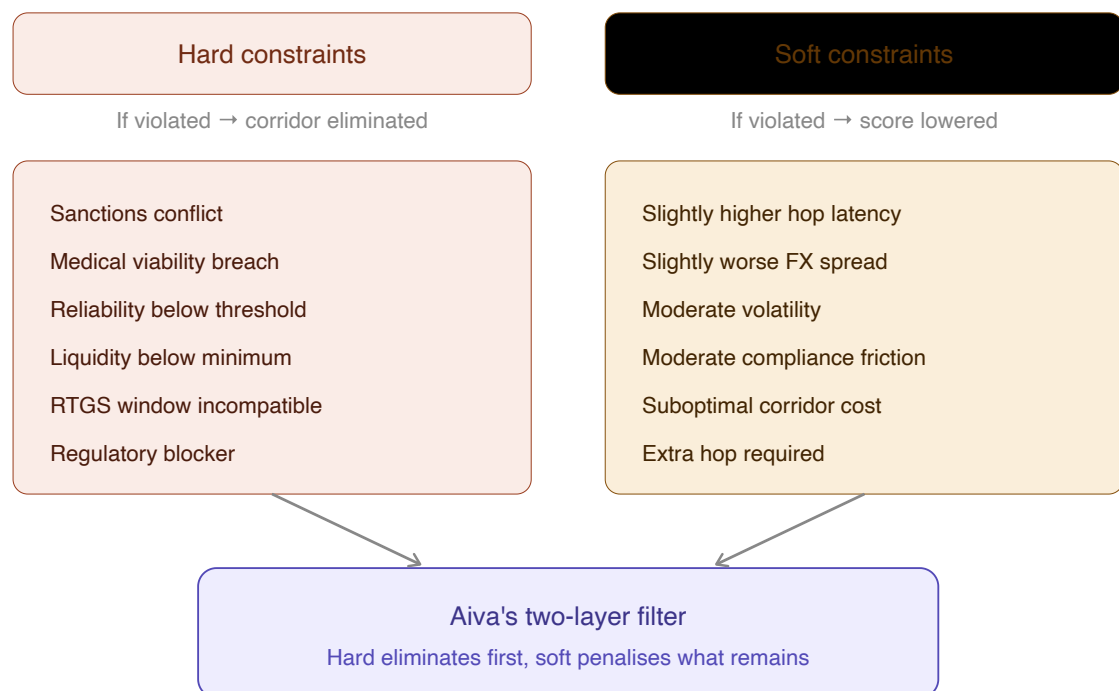
Multi-Constraint Optimisation — Aiva's Deterministic Decision Core

22 Apr 2026 (catch-up session)

If P2-Ch2 was about scoring, this chapter is about *solving* — turning scores into an actual optimisation problem with rules that must be obeyed.

KEY INSIGHT

Scoring alone isn't enough because Aiva faces contradictory objectives: fastest vs cheapest, safest vs most liquid, lowest FX exposure vs most compliant. You can't just average these — you need to optimise them simultaneously as a constrained global problem.



The optimisation tree: All corridors → Hard filter → Feasible set → Soft penalty layer → Utility optimisation → Optimal route. Scoring (from P2-Ch2) is now just one stage in a larger pipeline. The new front-end is the hard filter.

Formal model:

Optimise: $U(\text{route})$ Subject to: $H(\text{route}) = 0$ (hard constraints satisfied)

Where U is the composite utility function from Ch2 ($\alpha_1 \cdot \text{Liq} + \alpha_2 \cdot \text{Lat} + \alpha_3 \cdot \text{FX} + \dots$). H is the set of hard constraints. Standard constrained optimisation in mathematical form.

Volatility summed across hops:

$\sigma_{\text{total}} = \sum(\sigma_i \times \sqrt{t_i})$ – the $\sigma \times \sqrt{t}$ formula summed across every hop

A 3-hop corridor has 3 separate exposure calculations summed together. If σ_{total} breaches a hard cap, the entire corridor is eliminated.

COMPREHENSION CHECK 1

The book separates constraints into "hard" and "soft." Why is this distinction important? What would go wrong if Aiva treated every constraint as hard, or every constraint as soft?

CHRIS'S ANSWER

If every constraint was treated hard or soft, either the corridors will be eliminated or the scores would be lower.

Right shape, but the consequence matters more. All-hard = Aiva eliminates nearly every corridor, ending up with no feasible options (one minor issue kills good routes). All-soft = Aiva might select corridors that literally cannot execute (failed sanctions screening still scored and picked). The distinction is safety vs flexibility: hard prevents impossible routes from being selected, soft keeps good routes usable despite minor issues.

COMPREHENSION CHECK 2

V0 works with three pre-defined feasible routes. Does the hard filter matter for Singapore? Why or why not?

CHRIS'S ANSWER

The hard filter does not matter for Singapore trip as I am still experimenting, and it will always evolve with time.

Correct. Your honest pitch answer: "V0 uses three pre-feasible routes for the demo. In the full system, the hard filter sits in front of the scoring engine and eliminates impossible routes before they're evaluated. This is covered in P2-Ch4." Credible and confident — shows you understand the gap rather than hiding it.

V0 CONNECTION

V0's scorer.py implements the soft penalty layer. V0's selector.py picks the highest utility route. V0's three simulated routes are all pre-feasible, which is why the hard filter is skipped. A future enhancement would add feasibility_check() in router.py to filter infeasible routes before scoring.

PART III — CHAPTERS 2 + 3

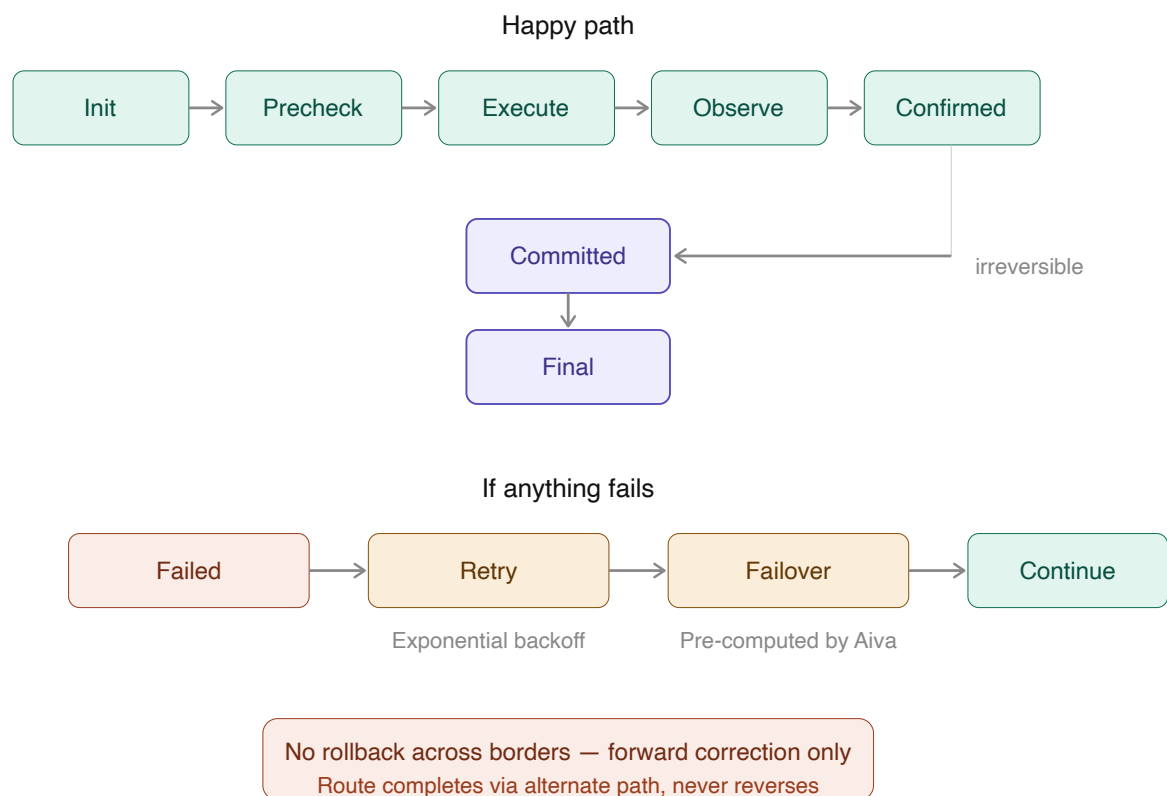
Execution Pipeline + Settlement, Retries & State Machines

23 Apr 2026 (combined session)

These two chapters are really one topic split in two. P3-Ch2 describes the pipeline (how hops flow), P3-Ch3 describes the stability machinery (how it stays reliable when things break). Together they explain how Aiva's decisions become real-world movement.

KEY INSIGHT

Aiva computes the brain decision. Rail is where failures actually happen: hops misbehave, correspondents delay, settlement freezes, handoff errors, timing windows missed. Rail exists to guarantee deterministic movement, reproducible behaviour, no ambiguity in hop transitions, and reliability higher than all legacy rails.



The hop state machine: Every hop flows through **Init** → **Precheck** → **Execute** → **Observe** → **Confirmed** → **Committed** → **Final**. Your V0's `executor.py` uses a simpler version: **initiated** → **scored** → **route_selected** → **executing** → **completed**. Same idea, simpler states. Critical property: there are no ambiguous states, ever.

The atomicity principle: "A hop either commits fully, or the system compensates. There is no in-between." No half-settled corridors. No partial transfers. Your `advance_state()` function enforces this with `valid_transitions` — that's the atomicity principle in code.

Idempotency:

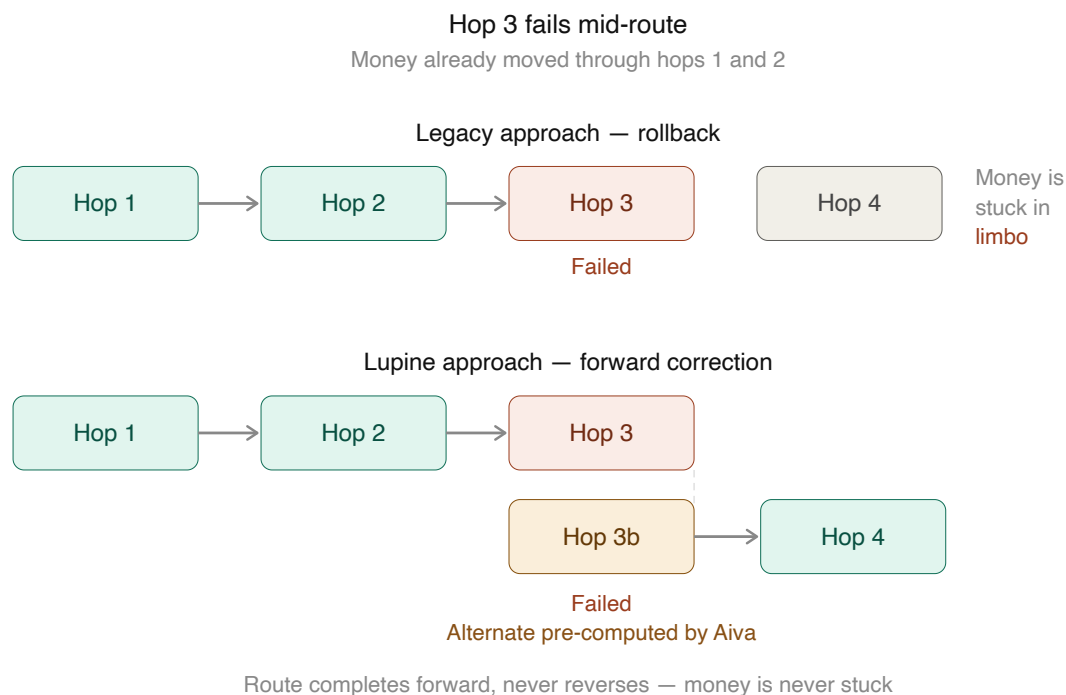
```
IdempotencyKey = SHA256(payment_id + hop_id + sequence_id)
```

Duplicate instructions are recognised and ignored. Prevents double-settlement, double-custody, replay attacks, accidental duplicate hops.

Retry with exponential backoff:

```
Retry(n) = BaseDelay × (GrowthFactor^n), GrowthFactor < 2
```

Retries include adaptive risk escalation (lower thresholds), time-window sensitivity (defer to next RTGS window), and liquidity feedback adjustment. Not just "try again" — smart retry logic.



Why forward correction, not rollback? Rollback across international borders is essentially impossible — once money leaves one institution, that institution doesn't control it. "Rolling back" would require a new payment going the other way, which is itself subject to all the same failure modes. Lupine pre-computes fallback routes (Aiva gives Rail a fallback tree with the execution blueprint), so when hop 3 fails, hop 3b is already ready. The route completes forward through a different path, never reverses.

COMPREHENSION CHECK 1

Why is rollback fundamentally different across international borders versus inside a single bank? Why does this matter for the Singapore pitch?

CHRIS'S ANSWER

Rollback is different across international borders because this would be a payment gateway in itself if the funds transferred are stuck mid flight.

You've got the right intuition. Sharpened version: inside a single bank, rollback is just a ledger entry reversal — they control both sides. Across borders, every hop involves a different institution, and once money leaves one, that institution no longer controls it. A "rollback" becomes a chain of new payments going the other direction, each subject to its own failure modes. Forward correction is simpler: complete the journey via a different final hop. Pitch version: "Legacy systems try to rollback failed payments, which turns one stuck payment into two stuck payments. Lupine pre-computes fallbacks so the money always completes forward."

COMPREHENSION CHECK 2

V0 enforces valid state transitions — you can't skip from initiated to completed. How does this map to the atomicity principle?

CHRIS'S ANSWER

I don't know the answer to the second one.

Atomicity = "either fully commit or compensate, never in-between." Enforced transitions make each state a promise: by the time you reach completed, every prior state has been visited and logged. If V0 allowed arbitrary transitions, you could claim a payment succeeded without running execution logic — a ledger lie. Your `advance_state()` raising `ValueError` on invalid transitions is the atomicity principle in code. Pitch version: "Every Lupine payment moves through a strict state machine. You can't skip steps, you can't be in two states at once, every transition is logged. This is what deterministic looks like at the execution layer."

V0 CONNECTION

V0's `rail_lite/executor.py` already implements the state machine, valid transitions (atomicity in code), state history tracking (observability), and UUID tracking (idempotency foundation). Missing: retry logic, failover, forward correction — because V0 routes don't fail. Pitch answer: "V0 demonstrates the deterministic state machine. Retry and failover logic is covered in P3-Ch3 and would be added in the full production system."

PART IV — CHAPTERS 1 + 4

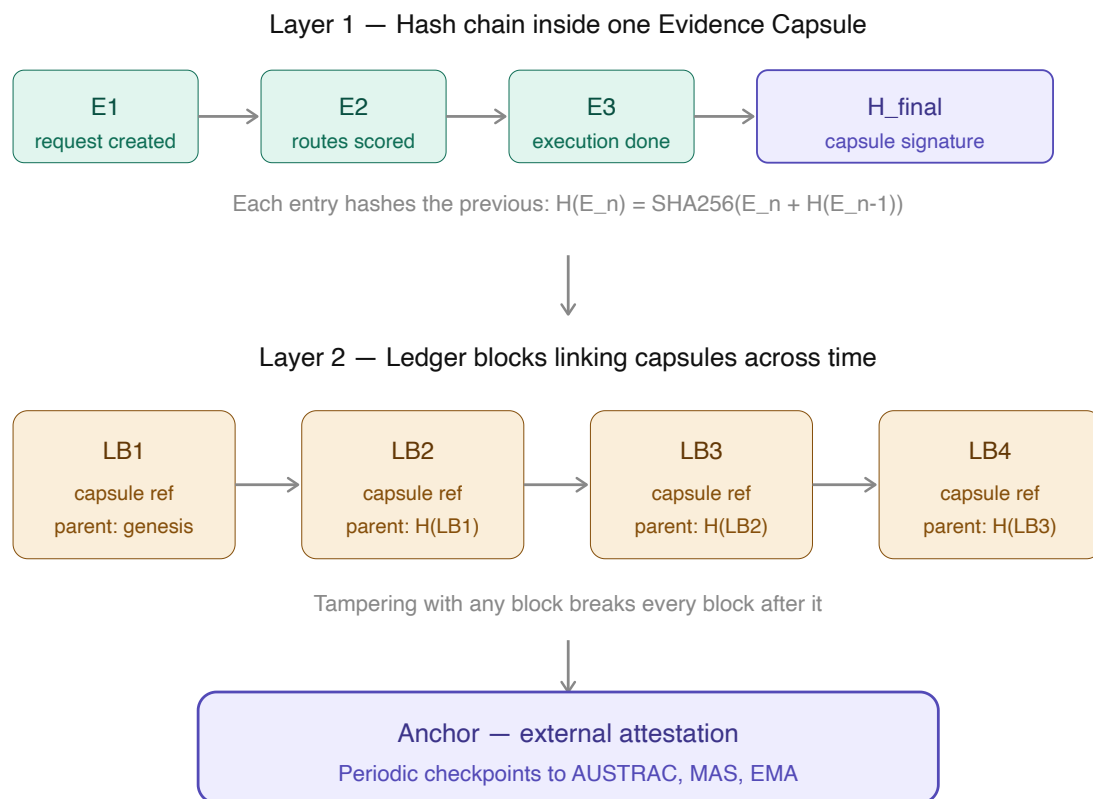
Cryptographic Evidence Chains + Audit Capsules

26 Apr 2026 (final theory session)

This is the Cloked layer — the trust primitive of the entire system. Aiva proves *why* a route was chosen. Rail proves *how* it executed. Cloked proves *that it actually happened, exactly as claimed, with no tampering*. Without Cloked, the first two are just unverifiable claims.

KEY INSIGHT

Banks and hospitals currently rely on logs that can be edited, deleted, or fabricated. SQL tables, S3 buckets, PDFs, handwritten slips. None of this is cryptographically accountable. Cloked makes claims mathematically verifiable.



Nested immutability — two layers of cryptographic linking:

- Layer 1 — Hash chain inside one Evidence Capsule.** Every event during a single payment (request created, routes scored, execution completed) gets hashed and linked. Your `cloked_lite/logger.py` implements this — `append_evidence()` chains the previous entry's hash, `verify_chain()` walks the chain detecting tampering.

- **Layer 2 — Ledger blocks linking capsules across time.** Each completed payment produces an Evidence Capsule, and those capsules get wrapped in ledger blocks linked across multiple payments. A chain inside a chain — capsule chain proves a single payment was honest, ledger chain proves the entire history is intact.

The tamper-detection mechanic: `verify_chain()` recomputes each entry's hash from its own data. If someone tampers with entry #2, the recomputed hash no longer matches the stored hash. Even if they fix that, entry #3's `previous_hash` reference is now broken. To tamper with one entry, you'd have to rewrite every entry after it, all the way to the end. Single-point edits are mathematically impossible.

Why Cloked is not a blockchain: Regulated industries can't use blockchain — evidence must be private, latency must be under 10ms, no need for decentralised consensus. Cloked uses the best parts of blockchain (cryptographic linking, append-only, tamper-evident) without the baggage (public ledger, consensus latency, gas costs).

Audit replayability: Given any block, regulators can retrieve the capsule, recompute the hash chain, verify the parent hash, replay the execution, rebuild the route matrix, re-evaluate compliance. Mathematically provable audits — not "we have logs" but "we can prove."

Dual-payload property (bridge to Phase 2): The same ledger handles both money (FX path, liquidity, settlement) and matter (cold-chain telemetry, custody chain, viability data). Same cryptographic guarantees, different schemas. The trust layer that proves money moved correctly will one day prove an organ arrived viable.

COMPREHENSION CHECK 1

Walk me through what would happen if someone tried to tamper with entry #2 in your evidence log. What does `verify_chain()` actually detect, and why does the chain break?

CHRIS'S ANSWER

Integrity should not be broken, any break in the chain means it has been tampered.

Right idea, but the mechanism matters because it's what makes the demo impressive. `verify_chain()` recomputes the hash for each entry from its own data and compares to the stored hash. If entry #2 is tampered, recomputed hash fails. Even if the attacker fixes entry #2's hash, entry #3's `previous_hash` still references the OLD hash, so the chain link breaks. To repair that, attacker must rewrite #3, then #4, then every entry. One tampered entry forces rewriting the entire chain after it. Pitch version: "If anyone tampers with any payment record, the cryptographic chain breaks at exactly that point and every record after it. You can't edit one entry — you'd have to rewrite history."

COMPREHENSION CHECK 2

The book says Cloked is "verifiable trust, not verbal trust." What's the actual difference, and why does this matter for someone sending AUD 500K to Singapore?

CHRIS'S ANSWER

The difference is that trust should be verified, and it should be displayed.

Right direction, but the consequence is what matters. Verbal trust = "we say it happened this way, here are our logs." Recipient must trust the sender's word, database, compliance team. Verifiable trust = "here's the evidence, you can mathematically verify it without trusting me at all." Recipient runs verification themselves. Even if the bank disappeared, the proof would still hold. You don't need to trust the source — the math does the trusting for you. For AUD 500K to Singapore: prove to your auditor the right route was selected, prove to MAS/AUSTRAC compliance was checked at every hop, prove the payment moved exactly as claimed. None of this is possible with current bank logs. Pitch version: "Banks ask you to trust their records. Lupine gives you records you can verify mathematically. You don't need to trust me — the cryptography does."

V0 CONNECTION

V0's cloked_lite/logger.py implements Layer 1 perfectly: SHA-256 hashing, chain linking via previous_hash, integrity verification through verify_chain(). Missing: Layer 2 ledger blocks across capsules, regulator anchoring, signature verification. For Singapore demo: V0 demonstrates a working evidence chain that proves any single payment was deterministic and untampered. Pitch answer: "The capsule chain is fully implemented; the cross-capsule ledger and external anchoring are covered in P4-Ch4 and would be added for production."

THEORY PHASE COMPLETE

All 11 V0-essential chapters mastered

26 Apr 2026

You now understand every layer of your V0 code at the conceptual level, and you have pitch language for every component. The book is no longer abstract — it maps directly to the code you can run on your laptop.

WHAT YOU CAN NOW EXPLAIN END-TO-END

The legacy problem (Ch1-2). Why messaging is the real control layer (Ch3). How corridors and FX exposure work (Ch4). How Aiva sees payments through 7 simultaneous graphs (P2-Ch1). How the composite utility function scores routes (P2-Ch2). Hard vs soft constraints (P2-Ch4). The state machine, atomicity, and forward correction (P3-Ch2+3). Cryptographic evidence chains and verifiable trust (P4-Ch1+4).

What's next: Run V0 locally, demo end-to-end, then pitch prep. 20 days until Singapore. Plenty of time.

PART II

External Readings

Primary-source passages from three public fintech prospectuses, reproduced in full and each prefaced with a short note on how it connects to the theory above. These are external documents — read them as evidence from the market, distinct from the study notes.

Wise IPO PROSPECTUS · 2021

WISE 2021 pp. 49–51

Our Infrastructure

WHY THIS MATTERS

Wise builds its own rails — “stitching together local payment systems, eliminating intermediaries” rather than leaning on correspondent banking. This is the real-world counterpart to the corridor engineering and deterministic routing you studied in Part I Ch4 and Part II Ch1: a leading operator deepening direct local integrations to collapse cost, delay and opacity.

Our Infrastructure

In order to solve the structural problems in the cross-border payments market, we realised we needed to replace the outdated correspondent banking system. Our infrastructure does this by stitching together local payment systems, eliminating intermediaries and manual processes, and hence removing the high costs, lack of transparency, delays and inconvenience that come with it.

Our infrastructure comprises four elements: (1) expansions and integrations; (2) technology; (3) regulatory; and (4) operations and customer support.

Expansions and Integrations

Our expansion journey begins with entering new geographies and partnering with local financial institutions to access domestic payment rails. Once we establish our presence, we deepen our connection by seeking our own regulatory licences, further integrating with local payment systems and rolling out features along the way. This is all carried out in a highly localised manner that is bespoke to each country, requiring close collaboration with local financial regulators. Our cross-functional teams work autonomously on many projects across regions in parallel, allowing us to increase our execution time and speed of expansion. The graphic below illustrates our geographical coverage as set out by the level of local integration and the type of available features.

Ten years ago, we started by offering two-way payments between the UK and Eurozone.

Today we have direct integrations with four local payment systems, partnerships with 85 local financial institutions, local processing in 88 countries, local account details in ten currencies and card issuance partnerships with Mastercard and Visa, and a cloud-based global connection with VisaNet.

Our industry-first integrations with local payment systems allow us to operate at faster speeds and lower costs. For example, by connecting directly to the FPS, we've been able to reduce our partner bank fees in the UK nine-fold and increase the speed of transfers to less than 20 seconds on average, which together has allowed us to drop our prices for customers sending to and from GBP. Similarly, when we integrated with Magyar Nemzeti Bank (the central bank of Hungary), customers sending to a Hungarian bank account saw a price drop of 14% on average and 82% of transfers to Hungarian Forint became instant (up from 17% in the third quarter of 2020). In Europe, we are an addressable BIC holder, enabling us to submit and receive payments within SEPA. In Singapore we are directly connected to FAST.

Technology

Our infrastructure is powered by modern, purpose-built technology, allowing us to scale rapidly whilst maintaining the quality and convenience our customers

have come to expect. We have over 500 engineers, distributed in six regions across the globe, constantly evolving and deploying our technology through over 90 production deployments per day (approximately 3,000 per month).

Our technological interface, which includes our website, mobile applications and third- party integrations, are built on the same open API, creating a seamless customer experience on any device or platform, anywhere in the world.

Our entire product suite is underpinned by a core engine of advanced payments technologies, including local payments systems integrations, a global real-time treasury management system, a smart, multi-currency ledger and automated KYC and AML verification and financial crime fighting.

Local payment system integrations are complex multi-step journeys, which usually start with integrating a local financial institution or an aggregator through a single API, and are broadened to include multiple API integrations to add partner redundancy. To deepen our technical integrations, we deploy physical data centres to connect with local payment networks where required by law.

Our proprietary global treasury management system provides real-time usage and liquidity predictions, smart fund routing, accounting and foreign exchange and financial risk management. This system gives us a global view of all liquidity flows and requirements throughout our entire network, allowing us to make efficient liquidity investment decisions. For example, the system enables us to cycle our funds around the world to meet different local payment system cut-off times and improve the speed of pay-outs across the network. Using this data set, we've built proprietary machine learning algorithms to help predict our customers' need for various currencies. Today, approximately 50% of traded volumes are predicted, enabling us to operate with low levels of working capital and keep our prices low.

We have also built a proprietary financial crime fighting engine which uses machine learning extensively to catch suspicious transactions and support our financial crime team. Given the magnitude of data points we process, our machine learning engine enables us to identify and prevent fraud while supporting instant SLAs. Combined, our financial crime team and KYC team

comprise over 450 Wisers working across verification (KYC), fraud, AML, enhanced due diligence, complaints, chargeback, requests for information and product compliance. These automated functions allow the teams to process over 21,000 ID checks per day.

Together, our technology delivers a scalable and highly resilient architecture with a 99.9% unadjusted uptime. Our infrastructure is cloud hosted and built to scale horizontally as our volumes grow. Our cloud-based operations also allow us to expand into new regions rapidly if our expansions require us to. Today we operate with 5x headroom based on current average utilisation, allowing us to continue to grow our business.

Regulatory

Our operations are supported by global licensing coverage. We hold 63 licences, across 12 different countries, supporting over 2,500 foreign exchange routes. We're on pace to keep increasing the number of licences, countries and currency routes we support. In FY2020 and FY2021, we added five licences, including in the UAE, Malaysia and Brazil, extending our currency routes across the Middle East, Asia and South America.

Our compliance team collaborates with regulators and law enforcement to access resources and tools, improve monitoring and keep pace with the latest local law developments. We have zero tolerance for the loss of any licences or public issuances from regulators that may inhibit our ability to serve our customers. As a result, we have never been the subject of an enforcement action.

Operations and Customer Support

Moving money should be stress-free, no matter how far it is travelling. We know convenience is central to our customers, so we've developed a hybrid model to automate

Take Rate

WHY THIS MATTERS

The disclosed economics behind the FX margin Aiva optimises. Total take rate vs cross-currency take rate is precisely the spread your FX scoring engine reasons about (Part II Ch2) — here grounded in three years of audited figures.

single upfront fee per transaction, consisting of a fixed and variable amount. For example, a typical transfer of £1,000 sterling into euros would have a fee of £3.69, consisting of a £0.20 fixed and 0.35% variable amount.

The amount of both the fixed and the variable portion of the fee depends on a number of factors, including the currency route, the transfer size, the type of transaction being undertaken and the payment method used.

We aim to price our products, services and features sustainably without cross-subsidising between customers, products or routes. Our fees are set by adding a sustainable margin to our underlying unit costs on a transaction level. We lower prices where and when we can sustainably reduce costs, for example, by directly integrating with local payment networks or by renegotiating lower unit costs. This is a major difference between Wise and many other businesses. As an example, when we integrated with Magyar Nemzeti Bank (the central bank of Hungary), customers sending to a Hungarian Forint bank account saw a 14% price drop on average as our costs of processing transfers to Hungary decreased. We have a team dedicated to understanding and reviewing the price of each of our more than 2,500 currency routes and other services. We have seen that, as volumes for a specific currency route grow, we are able to reduce our costs per transfer and therefore lower our average fees, while maintaining our margins. However, whilst we lower prices when our costs go down, sometimes our costs, and therefore our prices, can also increase.

A majority of our revenue is made up of fees on international transfers. Additionally, we also generate fees from other customer activities. This includes fees from same currency transactions, interchange fees from spend on our debit cards and other fees from product features related to the adoption of the Wise account.

Our total take rate is calculated as total fees across all customer activity as a proportion of volume, and was 0.77%, 0.73% and 0.66% for FY2021, FY2020 and FY2019, respectively. Our take rates have increased in line with the increase in the cross-currency take rate fees and the growth of other fees, driven in part by the introduction of the Wise Account.

The following table sets out our total take rate, cross-currency take rate and other fees as a proportion of volume for the periods indicated.

	2021	2020	2019
Total take rate	0.77	0.73	0.66
Cross-currency take rate	0.70	0.68	0.64
Other fees	0.07	0.05	0.02

Take rate as a proportion of volume (%), year ended 31 March.

Cross-Currency Take Rate

Our cross-currency take rate, calculated as the total fees on international transfers as a proportion of our volume was 0.70%, 0.68% and 0.64% for FY2021, FY2020 and FY2019, respectively.

This increased in FY2020 following a price increase for our customers as our unit costs increased, primarily from accelerated investments in our underlying infrastructure such as cloud storage.

In FY2021, we lowered prices for currency routes used by 76% of our customers, with a price decrease of 15% on average, and in a few cases up to 30%.

Furthermore, in FY2021, we introduced volume-based tiered pricing, allowing customers sending large monthly volumes to benefit from lower fees.

Despite this, our average cross-currency take rate increased in FY2021. Our fees depend on a number of factors including the currency pair, transfer size and payment method used. Certain currency routes are more expensive to operate due to higher local bank and partner fees, higher foreign exchange costs and tax costs. In FY2021, we experienced relatively higher customer activity on more

expensive routes. This increased our average cross-currency take rate and offset the impact price drops had on our take rate.

The price we report to our customers in our quarterly mission updates is based on a fixed basket of representative currencies which reduces the effect from route mix and other factors, making it a more accurate representation of our progress in reducing the cost of international transfers.

The following graph compares our cross-currency take rate against the price we report to our customers.

This shows that after increasing in FY2020, our customer price has been slowly decreasing, and highlights that lowering prices sustainably is hard and will take us time.

WISE 2021 pp. 30–33

Letter From Our CEO and Co-Founder

WHY THIS MATTERS

The founding problem statement — money lost to hidden fees and mispriced exchange rates — that motivates the entire legacy-problem thesis in Part I Ch1–2. Read it as the “why” underneath the architecture.

LETTER FROM OUR CEO AND CO-FOUNDER

Where we started Ten years ago, Taavet and I started Wise. We were both sick of losing money to our banks.

I was sending money home to Estonia every month with my bank and losing thousands to hidden exchange rate markups. I didn't like this then and I still don't today.

We came up with what we thought was a pretty clever solution. Every month, we looked up the actual exchange rate and then Taavet put his Estonian kroon into my bank account in Estonia and I topped up Taavet's UK account with pounds.

No fees, no bad exchange rates - it felt like we had outsmarted a system that was built to fleece us. It felt good.

We thought we might be able to help others save some money who'd been overcharged and underserved by banks. We built a simple website, crowdsourced a name from our friends, and Wise was born.

Where we are today In the decade since, we've grown from a simple money transfer service into an international account that helps a community of over 10 million like-minded people and businesses manage and move their money across the world.

Wise now moves over £5 billion across borders every month, saving customers more than £1 billion in fees a year.

Our Wise Account and debit card is replacing traditional international banking for millions of people and businesses. We've issued more than 1.6 million debit cards and customers held £3.7 billion in deposits as at 31 March 2021 across 56 different currencies in Wise accounts, and many use their Wise account numbers to get paid like a local in 30 countries.

Businesses now use us to manage their cash flow, pay suppliers, and get paid all over the world.

And by building this infrastructure for consumers and businesses, we've created a service over API that more than a dozen banks and enterprises use today.

Over the years we expanded to new countries, launched new features, and are still fighting as hard as ever against hidden fees. Throughout this, one thing has remained constant - our mission.

The Wise Mission

We are creating money without borders: instant, convenient, transparent and eventually free.

Money should move across borders instantly. Moving money across borders should be convenient. The fees should be transparent and not be hidden in exchange rates.

Eventually, we want to make money move across borders for free.

We're powering money for people and businesses: to pay, get paid, to spend, in any currency, wherever you are, whatever you're doing.

In 2011, this had never been attempted. In 2021, it is still a huge leap from where we are.

There remains a great deal of work to do.

Moving money into another currency is still a maze of hidden exchange rate markups, high fees, delays, and small print for most people. Sending, spending, receiving, and holding money internationally doesn't work like it should, because the international banking system was built for the past.

We're still in the early stages of making our mission a reality, but we know what we need to do and we have a team of over 2,400 dedicated people working to make it happen.

Making our mission a reality Today, moving money using Wise saves customers money, time, and hassle. It still mostly moves between bank accounts in different countries.

Increasingly, people and businesses use Wise to replace international bank accounts and hold their money, move money between their Wise Account balances, and spend it with their Wise debit cards.

Others are switching to banks, where cross-border transactions powered by Wise are built directly into their bank's apps.

As the world gets more connected, the need for moving money globally is only accelerating. We are making money work for that connected future.

The pages that follow lay out how we're going to do this. At its core, our plan revolves around:

- Building solutions to help personal customers manage their money internationally through the Wise Account.

- Making our Wise Business account work for businesses of all sizes so they can manage their international finances.
- Continuing to improve our infrastructure so that it can move money for anyone, anywhere and helping Wise Platform partners integrate Wise into their banks or apps. If we do these things right, we will make progress on our mission and save people and businesses much, much more than the £1 billion they currently save every year using Wise. What you should expect if you join us I like to tell our team that they haven't signed up to work at a company - but on a mission. Our shareholders should have the same mindset. Whether you own a small or large piece of Wise, you are not just buying a part of the company, but you're supporting our progress on creating money without borders. The process of becoming a public company is a milestone on our journey, but is not a defining feature in and of itself. The work that remains is making Wise cheaper, faster, and easier to use - all around the world. We've built Wise in a sustainable way to be financially independent - and will continue to do so. Our customers pay us enough fees - revenue - that lets us top up our balance sheet with profits for a rainy day. We are working on a difficult, structural problem that will take a long time to solve. For generations, banks have been defined by borders. Traditional bank accounts trap our money in one country, making international lives more difficult and expensive than they need to be. We are building a better way, but it won't happen overnight. Our mission has a long term horizon for success. It can take years to increase the speed of payments and drop fees by integrating into a nation's payment system. At the same time, we can create new features incredibly fast due to how our team and company is set up. Sometimes our progress may seem fast, but it requires patient, dedicated focus to make a lasting improvement in the fundamentals, such as the cost and speed. Shareholders should understand that we prioritise our team's efforts - and judge success - on the basis of mission progress. Revenue and returns are a function of our focus on our mission. We are not a company that prioritises short term solutions. If you're joining us as a shareholder, you should understand the long term nature of our mission and our focus on it. This is how we grew to help 10 million customers today and how we aim to help 100 million customers in the future. Staying true to our values Becoming a

publicly traded company is something that we've considered for a long time. We are approaching this evolution in our own way that reflects our mission and values. This moment in time is marked by three key characteristics. 1 We are taking steps to align ownership and mission by bringing customers into our shareholder base. Doing this ensures we are solving their problems the best we can. 2 We are taking steps to become public in a way that is transparent and fair. By doing a Direct Listing, we are ensuring everyone gets the same opportunity to own a part of Wise as large institutions. We're allowing our existing shareholders to sell on their own terms and allowing the open market to set a price in a transparent way that's fair for all. 3 Keeping our long term focus. We are putting in place a voting structure that ensures our team can focus on our mission over the long term. Join us At the time of becoming a public company, it's easy to get distracted by evaluating revenue and profits and debating a share price. But for us, this moment is about our customers finally being able to become owners of Wise. Money is an important part of our lives. But thinking about how to get it from one place to another should never be an expensive or inconvenient worry. Many of our 10 million customers live away from home. The people who use Wise to move money around the world every day are individuals supporting their families, parents paying school fees, and business owners funding their entrepreneurial dreams. When we talk about removing borders from money, what we're actually talking about is saving them money and making their lives easier. It's what we started doing 10 years ago and it's what we're going to be doing for a long time. If you believe in that, we're excited to have you join us on this mission. Onwards.

Adyen IPO PROSPECTUS · 2018

ADYEN 2018

pp. 63–65

An Overview of the Payments Value Chain

WHY THIS MATTERS

The canonical map of the payments value chain — issuers, acquirers, networks, gateways and the participants between. This is the structural terrain every routing decision in Lupine traverses; keep it beside the Part I messaging-stack diagram.

AN OVERVIEW OF THE PAYMENTS VALUE CHAIN

Consumer electronic payments can take various forms. These include credit cards, debit cards, pre-paid cards, and many local and other payment methods. According to Capgemini, card-based payments are the largest of the aforementioned categories based on transaction volume, with credit and debit cards taking the largest share globally.

The figure below provides an illustration of a typical electronic payments value chain.

Figure 1: Typical electronic payments value chain Payments systems typically involve a number of participants. In its most common form, the traditional card-based value-chain involves five parties whose interactions are governed by a card network. Below is an overview of the participants and their respective roles:

- ☒ Shoppers buy goods or services from merchants. These purchases can be made across a variety of channels. Traditionally, most transactions have taken place in-store (POS). However, with advances in technology, volumes are increasingly shifting towards the online and mobile channels;
- ☒ Merchants accept electronic payments in return for providing goods and services;
- ☒ Acquirers sign card acceptance agreements directly with merchants and are typically the merchants' primary point of contact, providing them with access to the card networks. The acquirer performs the necessary KYC and due diligence review of the merchant in order to assess and mitigate fraud and/or bankruptcy risk. Acquirers are also responsible for delivering funds, which the networks have received from issuers, from card networks to merchants, as well as routing transactions from various channels to the card network;
- ☒ Card networks set the rules concerning the processing of transactions. The largest and most well-known networks are Visa and Mastercard, but there are also a significant number of other international

players as well as local card networks. Card networks are the backbone of the electronic payments system. Their responsibilities typically include connecting and switching transactions between acquirers and issuers, enabling electronic payment authorization, as well as clearing and settlement. According to The Nilson Report ("Nilson"), as of April 2018, the six largest card networks were Visa, Mastercard, UnionPay, American Express, JCB Credit and Discover, together generating purchase volume of \$23.0 trillion for 2017, with disparities between geographies; and Issuers provide shoppers with credit and debit cards. Card networks route authorization requests to the issuers, who then authorize the merchant to accept the payment once necessary identity and compliance checks were performed.

Over time, there has been a trend towards additional participants joining the value chain:

- Terminal/gateway software providers give the merchant the necessary hardware or software (depending on the channel) to accept payments. This is intended to facilitate acceptance from the broadest possible range of payment types;
- Processors serve technical roles such as authorization, data transmission, data security and settlement functions as an outsourced service to acquirers. The terms "acquirer" and "processor" are often used interchangeably, but they are two distinct functions that, in some cases, are provided by the same entity; and
- Risk management providers try to assess whether the transaction is genuine or fraudulent on behalf of the acquirers.

Alongside Acquirers, the above can be collectively referred to as payment service providers ("PSPs"), and typically PSPs perform one or more of the above functions.

Role of Adyen in Value Chain

Adyen, with its single, integrated platform, operates across the entire value chain and offers simplicity for merchants by eliminating the need to work with a disparate group of gateways, risk management providers, processors, and acquirers.

Figure 2: Illustrative Adyen value chain Overview of the mechanics of a transaction A typical card-based payment transaction consists of the following

steps which can be grouped into two sub processes; authorization and settlement. In the first part of the transaction flow, authorization, the following steps take place: 1. The shopper presents / inserts card details at the POS terminal / online gateway, where in case of a POS terminal the account information is then captured from either the chip, magnetic stripe or, in the case of a mobile wallet, virtually through the use of for example near-field-communication (NFC) technology or a QR code; 2. The transaction data is then routed to the acquirer processor through an internet connection or online gateway; 3. The acquirer processor / risk manager performs the relevant initial risk management checks and then routes the transaction to the card network (e.g. Visa, Mastercard, American Express); 4. The card network identifies the relevant issuer and routes the transaction to the issuer processor; 5. The issuer processor validates the identity of the cardholder and provides authorization to the card network; 6. The card network forwards the authorization note to the acquirer processor; 7. The acquirer processor then sends the authorization note to the POS terminal / online gateway, enabling the merchant to proceed with the transaction and providing the shopper with a confirmation and a receipt.

The second part of the transaction, settlement, occurs at a later point and involves the acquirer and issuer exchanging purchase information to complete the transaction: 1. The merchant will send a capture request to the acquirer processor (via the POS terminal / online gateway); 2. Once the acquirer processor receives these captures, it routes the capture to the respective payment network; 3. This is then transferred to the applicable card issuing bank which informs the customer of their transaction by posting it on their statement; 4. The payment network then calculates net settlement positions for the acquirer processor and the card issuing bank, advises both parties of the amount, and submits a transfer order to a settlement bank; 5. The settlement bank acts as a facilitator for the flow of funds between the acquirer processor and the card issuing bank; 6. The acquirer processor transfers the received funds into the respective merchant's account.

Fee structures for value chain participants On a simplified basis, fees are traditionally charged to the merchant as a percentage of the value of each transaction. The fees for various players include: ☐ Processing fee – charged to

the merchant by the gateway or POS payment provider; ☐ Acquiring markup – charged to the merchant by the acquirer; ☐ Interchange fee – charged to the acquirer (via the card networks) by the issuer and usually passed on to the merchant. The card issuer takes on the credit risk of the shopper. The interchange fee has traditionally been the largest part of the total fee charged to the merchant; and ☐ Card network fee – charged to the acquirer by the card networks and passed on to the merchant.

This is typically much lower than the interchange fee but is charged on all transactions which occur over the relevant network, Adyen receives the processing fee and acquiring mark-up. The Interchange fee and card network fee are costs for Adyen which are usually passed on to the merchant.

Technology is having a significant impact on the traditional card-based payments systems with innovation happening across the value chain. A number of new form factors and device types are now being used to initiate electronic transactions at merchant locations including e-Wallet enabled devices such as Apple Pay and Google Pay. In addition, POS terminals and systems are becoming increasingly sophisticated with rich functionality beyond payments and multiple customer interfaces such as QR codes, NFC and Bluetooth communication. Additionally, technologies such as data analytics and machine learning are increasingly being used by industry participants to combat fraud, identify shopper patterns and provide insights to merchants and consumers. New tokenisation technology is also allowing for more seamless consumer experiences such as in-app and recurring payments as well as "one-click" and "zero-click" functionality.

New technology developments are also resulting in the proliferation of local payment methods especially in emerging markets where cash has historically been more prevalent. Given limited investment in traditional card-based payments infrastructure, many countries are more amenable to adopting alternative payment methods in addition to card-based payments. The rapid rise of Alipay and WeChat Pay in China and Paytm in India are examples of such fast growing local payment methods.

On the other hand, card-based systems have been around since the mid-20th century and have been quite resilient to evolving technologies. The infrastructure benefits from the trust gained from businesses and consumers, obtained over its long history of global operations, the tremendous scale of transactions processed, and well-established rules of processing and exception handling.

Adyen is able to offer both card-based payments and local or alternative payment methods to its merchant base and benefits from the growth of both card-based and local or alternative payments.

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One Single Platform Built for Growth

WHY THIS MATTERS

Adyen's argument for a fully-integrated single platform — one stack from gateway to risk to processing to settlement — directly parallels Lupine's unified routing brain versus the stitched-together legacy systems described in Part II Ch1.

Strengths

Adyen's competitive strengths are underpinned by its fully-integrated proprietary platform, which allows Adyen to support merchant growth around the world and across channels, including through data-centric solutions that aim to increase revenue and by developing new features and services on its single platform alongside merchants in a co-creation process. This has made Adyen a trusted partner of many of the world's most recognizable companies. Adyen is supported by an experienced management team that actively fosters an entrepreneurial culture.

One single platform built for growth The typical payments landscape has been characterized by a fragmented patchwork of providers and legacy systems, which Adyen believes leads to an inferior shopper experience, both explained

and unexplained declined authorizations, low conversion rates and a high number of fraudulent transactions leading to administrative costs for merchants. In this context, the Adyen team set out to fundamentally change the payments industry by building a single, fully-integrated global platform which seeks to provide a high- quality level of service to merchants.

Illustrative Traditional Value Chain

Illustrative Adyen Value Chain

Full end-to-end payments stack Adyen's state-of-the-art platform integrates the full payments stack. The Adyen platform was built in-house and designed to replace several blocks of the payments value chain with a single, integrated platform.

Where traditional platforms provide for separate gateway, risk management, processing and acquiring solutions, Adyen combines all these functions into one platform. This integrated approach provides Adyen with better intelligence into the payments flow, allowing it to provide high-quality service levels and transparency to merchants. Furthermore, Adyen's platform leverages best-in-class technologies including machine learning, data mining and artificial intelligence.

Adyen's full payments stack provides merchants with a holistic view of the payments flow through a single integration and one contract, including insights into why a specific transaction is rejected by an issuer.

These insights can help to increase sales for merchants while improving the experience for shoppers, thereby contributing to higher revenues and lower costs. This in turn helps to drive Adyen's merchant retention and acquisition. Further, the one contract approach allows for increased efficiency when it comes to the merchants' reporting and reconciliation.

Technical Superiority

Adyen's single platform also has significant benefits for Adyen in terms of ease of management, operational efficiency, robustness and agility in quickly launching

new features. The platform has been fully developed, and is run, in-house, and all new features have been designed with the ability to scale in mind. Adyen's proprietary software for its platform runs on open source technology and remains largely independent of software commercially licensed from third parties.

Adyen believes that not being limited by third party software in its core operations has been a key factor behind its ability to rapidly increase the number of transactions that the platform can process. Adyen also believes that managing all operations in-house helps to maintain a high level of security and performance.

The end-to-end control further enables flexibility for Adyen to combine platform components in different configurations to meet the various needs of its merchants and allow for scalability.

Delivering significant benefits for merchants Adyen's proprietary payments stack enables Adyen to deliver high-quality service levels to its merchants, providing them with: A global platform with local depth Adyen offers its merchants access to payments worldwide through one platform and a single integration from the merchants' perspective. Given the proliferation and importance of local payment methods, Adyen is focused on providing a comprehensive, global payments solution to its merchants. Through its single platform, Adyen offers access to a large number of global card schemes (e.g. Visa, Mastercard) and local payment methods tailored for shopper payment preference in each market, such as in Latin America (e.g.

Boleto), Asia (e.g. Alipay) and Europe (e.g. iDeal). The company also offers local acquiring on a global scale. The use of local acquiring and payment methods generally increases authorization rates, reduces foreign exchange rates and foreign transaction fees, lowers interchange and card network fees, and reduces payout time per payout currency for the merchant.

Unified commerce across all channels Adyen believes that the future of payments is 'unified commerce', meaning, more generally, the elimination of distinctions between the channels through which a shopper interacts with a merchant and, in the case of Adyen's current product offering, combining POS,

online and mobile standardized across channels for merchants. As shoppers' purchasing journeys are becoming increasingly complex and shopper needs evolve to require ever-increasing convenience, Adyen recognizes the importance for merchants of having a unified payments solution. This is the case for multi-channel merchants (such as, for example, retail or hospitality) as well as single or dual-channel merchants (such as digital goods providers or grocery stores).

Traditionally, both payment service providers and merchants have been challenged with in-store and online/mobile payments running on different technology and systems. Adyen's platform has no such limitation, being built as a channel agnostic platform from day one. Adyen's platform provides the same back-end payment processing and settlement infrastructure for merchants whether their shoppers make purchases online, on their phones or in-store. Unburdened by the need to patch together disparate channel-specific systems, Adyen is able to offer merchants unified commerce solutions out of the box, which can be tailored to their needs, are aimed to ensure that merchants are ready for future trends in commerce and are able to quickly roll out new stores and enter new markets while retaining a centralized payments platform and holistic view of their shoppers, regardless of what channel they shop through.

Data-centric solutions to increase revenue while reducing risk Adyen's platform aims to optimize payments, reduce fraud and provide shopper insights to merchants.

Traditional payments systems are a so-called 'black box', whereby transactions are either approved or declined – with little insights as to why that happened, especially as payment service providers are only one part of the value chain. In contrast, Adyen's control of the complete payments stack results in a high degree of transparency, including as to why transactions are rejected. Adyen is able to leverage advanced technologies and provide intelligence and insights on how to adapt formats, user interfaces, and customer journeys to ensure the highest chance of transaction approval. Furthermore, Adyen's visibility into the payments value chain allows the Company to gather data on shoppers' behaviors, which can then be used to generate actionable insights for merchants on how they can best serve their shoppers and optimize their systems and settings to achieve higher authorization levels. This ultimately leads to a smoother

experience for the shopper without any compromise in terms of risk management and fraud detection. In sum, these insights can serve to increase the number of successful transactions for merchants while decreasing complexity for their shoppers, thereby supporting higher revenues and lower costs. This in turn helps to drive Adyen's merchant retention and acquisition.

Membership to ongoing innovation Adyen has invested heavily to provide the best-in-class payment experience for all of its merchants with a strong focus on the security, reliability, scalability and performance of each component of its platform. The partnership between Adyen and its merchants provides merchants with access to ongoing innovation, as Adyen continuously improves its products and has a regular software release schedule with improvements deployed every four weeks. This frequent release schedule ensures that merchants will benefit from prompt access to the latest developments. Adyen's aim is for any new feature development to be standardized and made accessible to all merchants.

Most innovations are a direct result of a close collaboration between Adyen and its merchants. To regularly improve its platform and meet its four-weekly release schedule, Adyen organizes its projects around goal-oriented, cross-functional teams, called 'streams', which are comprised of individuals from the business side and the development side of the company and built in collaboration with merchants. These 'streams' benefit from being agile and able to quickly implement their ideas and verify them in practice, allowing for fast launch and iteration cycles.

Trusted partner of many of the world's most recognizable companies From its outset, Adyen has focused on providing payments on a global scale, recognizing that merchants need to be able to seamlessly carry out cross-border commerce. Further, Adyen has consistently sought to provide these merchants with best-in-class service and transparent pricing. Adyen currently serves several thousand merchants with an offering of the most relevant payment methods, whether global or local, in numerous currencies across six continents and across a wide-number of industries, including retail, travel, digital services, hospitality and marketplaces. Adyen's merchant portfolio includes Uber, Netflix, Facebook, Spotify, Etsy, Vodafone, Sephora, Tory Burch, L'Oréal and booking.com.

In relation hereto, Adyen processed 3.7 billion transactions for its merchants in 2017, translating to €108.3 billion in processed volume in 2017, as compared to €66.3 billion in 2016. Over the period from 2015 to 2017, Adyen's annual churn rate was less than 1% of processed volume.

An experienced founder-led management team fostering an entrepreneurial culture Since Adyen's inception, its management team has focused on fostering an entrepreneurial culture, at the heart of which rests a commitment to superior service for merchants and to creating benefits for all stakeholders, as represented by the "Adyen Formula" (see "-People and Culture"). The principles embodied in the Adyen formula include building solutions to benefit all merchants (not just one), launching new products and solutions fast and iterating and making good choices in order to build an ethical business and drive sustainable growth for Adyen's merchants.

Adyen believes that the "Adyen Formula" is a key driver in enabling Adyen's ability to hire and retain top talent, to foster the Adyen culture and to have fun while changing the payments industry. Talent development, diversity, and the retention of Adyen's entrepreneurial culture are considered business imperatives to Adyen. Adyen's founder-led management team, with years of experience in the payments industry, has assembled a very strong core team that manages key business areas and regions. This team includes members who have worked together for nearly 15 years, as well as young talent, mainly trained in-house, and is supported by finance, operations, regulatory and other functional experts and payment and technology leaders. Following the completion of the Offering, the management team is expected to remain in place and members of the current management team will continue to hold significant equity ownership in Adyen.

A business that is difficult to replicate Adyen's business is difficult to replicate given the following characteristics: ☐ a single, globally integrated platform enabling a seamless and high-quality transaction experience for both merchants and shoppers; ☐ tech DNA and payments expertise – a "technology first" business from inception benefitting from its speed of innovation reinforced by quick time to market with new products; ☐ strong relationships with some of the world's largest and most globally connected merchants; ☐ a global presence that is reinforced by local expertise, partnerships with key constituencies and a suite of

acquiring and regulatory licenses and regulatory know-how each built up by Adyen over the course of the years and responsive to the various regulatory and commercial challenges in the payments industry; ☐ a powerful culture attracting some of the best people in the payments industry; and ☐ pricing transparency.

Strategy

Adyen operates in an agile manner, guided by the Adyen Formula and the ever-evolving needs of merchants. These principles – to help merchants grow, to change the payments landscape, and to have fun while doing it – position Adyen as a growth partner to the merchants that it serves. Adyen aims to be at the forefront when it comes to developing new functionality as business models evolve.

Adyen believes that it is well positioned to significantly grow its business and increase scale over the coming years, with growth opportunities mostly centered on the following three strategic pillars: 1) Continued Focus on Enterprise Merchants Adyen has traditionally grown its business by solving payments-related issues for global enterprise merchants, which Adyen generally defines as large, multinational corporates. Adyen still sees significant potential for upside in this category for the following reasons: ☐ Deepening relationships with existing merchant base: Adyen aims to grow its business within its existing merchant base, which includes some of the largest multinational merchants. This increase is expected to come through a combination of sources, including (i) Adyen's merchants' organic growth, (ii) a deepening of Adyen's product offering and (iii) an increase in Adyen's proportion of transaction volume with its current merchants, which often is achieved through an increase in geographies covered by Adyen with its current merchants and/or adding new channels for merchants, such as having a merchant add Adyen's POS services if it is already using Adyen for e-commerce transactions or vice-versa. ☐ New merchants: Acquiring new enterprise-level merchants across different verticals and geographies. Adyen has a dedicated sales force focused on acquiring new globally operating enterprise-level merchants in each of the geographies and industries where it is present. ☐ Capitalizing on evolution in business models: As Adyen's platform is able to

quickly respond to changing merchant needs, it is well positioned to benefit from its merchants' changing business models.

2) Increasing Adyen's Unified Commerce footprint

The retail industry is rapidly evolving, with merchants realizing that their ability to connect with shoppers across channels can have a significant impact on their businesses. Online and mobile retailers are increasingly establishing a physical presence and brick-and-mortar retailers are increasingly pursuing an online and mobile strategy. Increasingly, merchants aim to build various types of sales and distribution models across sales channels.

Adyen believes it is well positioned to capitalize on this paradigm shift in retail as result of the following: ☒ Holistic view of payments on Adyen's single platform: Adyen's single platform has been built for unified commerce from day one, with all transactions, regardless of channel, identified similarly in the merchant's system. This approach allows the collection of data that can deliver unique shopper insights and simultaneously helps to combat fraud. Adyen's back-end infrastructure for

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Marketplace Solutions & RevenueAccelerate

WHY THIS MATTERS

How a platform monetises value-added layers on top of core processing. A useful lens on where margin beyond the base take rate actually comes from.

Marketplace Solutions ("Adyen MarketPay")

Marketplaces (such as Etsy) are a fast-growing business model, providing opportunities for their users, or "sub-merchants", which include a whole new market of small business owners and entrepreneurs on the Internet.

Marketplaces are faced with unique challenges, being responsible not only for

integrating payments into their platform, but also the need to facilitate payouts of funds to their sub-merchants.

Consequently, marketplaces have a substantially higher regulatory burden, particularly in relation to KYC and on-boarding requirements for a large number of sub-merchants.

The combination of these forces – a growing market with unique challenges in payments and regulations – led Adyen to create Adyen MarketPay in 2016. Adyen MarketPay is a global and flexible payment and payout solution for marketplaces. This payments solution is an extension of the same full payments stack as the Adyen platform and includes automated and secured on boarding of sub-merchants, the ability to automatically split payments between merchant funds and sub-merchant payouts, optimized payout facilitation, and managed compliance. As a result, Adyen MarketPay is a "one-stop-shop" solution for marketplaces to offer their services globally. Merchants are able to create flows that make sense to their marketplace using Adyen's set of APIs, allowing for absolute flexibility.

Through Adyen MarketPay, regulatory and compliance burdens can be alleviated to a large extent by Adyen, which executes KYC and anti-money laundering checks in compliance with global regulatory requirements. This aims to allow sub-merchants to concentrate on servicing their shoppers while Adyen automates compliance and KYC-related tasks. While Adyen supports marketplaces in meeting their regulation and compliance requirements, the primary responsibility remains with the marketplaces.

Data-enabled products Due to Adyen's integrated platform, Adyen has visibility on payments data throughout the value chain, as well as infrastructure that can facilitate complex, real-time data analysis on that information. Adyen's data-enabled products include three main offerings: RevenueAccelerate, RevenueProtect and Shopper Insights.

Authorization rate optimization ("RevenueAccelerate") Due to the traditionally complex and fragmented payments landscape, payment systems have largely been a black box for merchants. Transactions were either approved or declined, without significant insight or room for optimization. RevenueAccelerate aims to

correct for these inconsistencies in the payments ecosystem by converting wrongly refused card transactions into approvals. RevenueAccelerate is a data-driven suite of automated tools, which work in the background of each payment to increase authorization rates and revenue.

Drawing on Adyen's direct access to card network data, RevenueAccelerate uses machine learning, data mining and AI techniques to build intelligent logic to adapt both the format and route of each payment request in real-time, targeting the highest chance of an approval from the issuing bank. RevenueAccelerate has a set of features including intelligent payment routing, dynamic card validation, auto-retries and account updater. All of these features work automatically and in tandem to provide merchants with better authorization rates. Importantly, since the ecosystem is dynamic, with entities across the value chain making changes and updates of their own, the entire RevenueAccelerate system automatically adjusts to changes as they are made.

Fraud Prevention ("RevenueProtect")

As with RevenueAccelerate, Adyen's RevenueProtect product uses machine learning, data mining and AI techniques to screen transactions for fraud. RevenueProtect performs a number of checks resulting in a total score for every transaction, which gives an indication of the likelihood of the transaction being fraudulent.

Unlike other fraud prevention systems, Adyen's merchants can configure scoring rules themselves, offering both the benefits of automation and data-driven logic, with the transparency and control that merchants expect. Further, for merchants working in industries that prefer manual review processes, Adyen offers a robust set of manual review interfaces and outsourcing services to assist in this regard.

Central to RevenueProtect is Adyen's "ShopperDNA", which is a proprietary transaction-linking algorithm.

ShopperDNA works in real-time to create a holistic view of related transactions across a merchant's entire history. This allows Adyen to identify fraudsters even as they change devices, identities and payment methods, as well as track loyalty and reduce false positives on valid shoppers who might, for example, use

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Key Benefits of the Single Platform

WHY THIS MATTERS

The data and operational advantages that compound from a single integrated platform — echoing the seven-graph data advantage that powers Aiva's scoring engines (Part II Ch2).

Additionally, Adyen's platform is well-adapted for emerging business models, illustrated by its success in providing payment services to Silicon Valley-based internet companies.

Adyen operates across industries – most notably in retail, travel, digital services, hospitality and marketplaces. For the year ended 31 December 2017, as measured by net revenue, Adyen's top 10 merchants represent 33% of net revenue and its top 120 merchants represent 69% of net revenue.

Measured by processed volumes, Adyen's top 10 merchants represent 39% of processed volumes and its top 120 merchants represent 83% of processed volumes. Adyen had 3,401 merchants billed in December 2017 (with 478 representing more than €1 million each in processed volume per month during the full year ended 31 December 2017), as compared to 4,510 billed in December 2016 (with 336 representing more than €1 million each in processed volume per month during the full year ended 31 December 2016) and 4,649 billed in December 2015 (with 225 representing more than €1 million each in processed volume per month during the full year ended 31 December 2015). Adyen's processed volumes have grown as the result of both growing volumes with existing merchants and adding new merchants. Adyen regularly monitors its merchant base and removes inactive merchants from the platform, which has led in recent years to a decline in the total number of merchants served but an improvement in the quality and sustainability of the merchant base and high growth in the number of merchants providing high volumes.

Partnerships

Adyen's main partnerships are with its acquiring and local payment partners, as described above in "Global Acquiring and Local Payment Methods". Other significant partnerships include its banks, the POS terminals provider that produces the terminals that Adyen provides to merchants using Adyen's POS solution, as well as warehouse and logistics service providers for the shipment and storage of these terminals.

Technology and Engineering

Adyen's development team began building the Adyen platform from the ground up twelve years ago and designed it to provide a best-in-class payment experience for its merchants with a strong focus on scalability, security and performance of each component of its platform.

Adyen's platform allows merchants to accept payments through all channels in a secure, efficient and flexible manner. By offering a single, quick integration without compromising on features, merchants can concentrate on growing their business rather than managing separate payment solutions for different channels.

Key Benefits of Adyen's Payments Infrastructure

1) Cost-efficient, scalable architecture: Adyen's platform was designed with scalability in mind. One of the examples of this is the fact that transactions can be processed over multiple independent servers – resulting in linear scalability of the platform. Due to this design, Adyen has been able to rapidly scale up in a very cost-efficient manner in the past and will be able to deal with growing volume in the future.

The platform has been entirely built in-house and to date remains largely independent from commercial third-party software providers, resulting in increased security and ease of maintenance. These benefits are further amplified by the modular architecture of the Adyen platform, which provides a high degree of flexibility for Adyen to select and further develop the best technology for individual parts of its service.

2) High resilience and low latency: Adyen leases server space in six data centers around the world (see "- Property Plant and Equipment") with high-performance servers, storage and computer networking capabilities, located across multiple continents and deployed in an "active-active" setup, which means that during normal processing, a payment request can be handled over Adyen's private network by the servers closest to the merchant and/or shopper, resulting in faster processing times. At the same time, transactions can be easily re-routed to another data center in the case of an outage of one of the data centers. This data center set-up reduces the platform's dependency on any single data center and mitigates against network, energy and natural disaster risks. All data centers pass annual comprehensive evaluations for physical access control, security monitoring and resilience.

3) Reconciliation capabilities: At the core of Adyen's payment platform is a double-entry accounting system, which accounts for all financial events (such as payments) and ensures that both credit and debit sides of bookings are balanced. Using double-entry bookkeeping, Adyen is able to obtain a high degree of automated reconciliation capabilities, benefiting both Adyen's and its merchants reporting capabilities.

4) Security, system monitoring and alerting: Adyen prioritizes security throughout its platform, operations and organization. Adyen makes conscious architectural, design and implementation choices to structurally address security risks. Adyen's staff is trained and is engaged in the ongoing security of the platform and for new developments. The platform continuously monitors activity, and anomalous data (also known as "exception data") is analyzed for indicators of unauthorized activity. Furthermore, Adyen's platform is compliant with Payment Card Industry Data Security Standards (PCI DSS) and other relevant PCI standards.

Technology Strategy

1) Merchant-driven innovation Adyen aims to develop generic functionality that will benefit all of its merchants. The source for most innovation generally comes from leading businesses in their respective industries, which Adyen then

develops platform-wide, thereby providing advantages for all merchants on the Adyen platform.

2) "Streams" - Collaboration between developers and commercial teams To continuously improve its platform, Adyen organizes its projects around goal-oriented, cross-functional teams, called "streams". These are comprised of individuals from the business organization (product specialists and functional experts in areas such as finance, compliance etc.) and the development organization (front-end and back-end developers).

This setup ensures that developers and product specialists collaborate closely on new product developments as well as on performance optimization, aiming to create the highest quality output for both Adyen and its merchants. The success of the streams is made possible by Adyen's autonomous culture that values a bottom-up approach, teamwork and the ability to communicate effectively and directly across teams.

3) Launch fast and iterate Adyen continuously improves its products and has a regular software release schedule with improvements deployed, typically on a four-weekly basis (whereas the industry norm is to release only a few times a year).

This frequent release schedule aims to ensure that merchants will benefit from immediate access to the latest developments.

Adyen has developed full testing capabilities with continuous integration, with advanced testing features including the in-house 3D-printed, POS test robots, which are used to remotely test the roll-out of new software on Adyen's POS terminals.

4) Investing in technology talent Adyen has invested extensively in its technology talent. As of 31 December 2017, Adyen had 34% of its employees in engineering and technology related roles, as compared to 33% in each of 2016 and 2015. The development teams often engage directly with merchants, supporting the commercial teams to improve the merchant's payments experience and implement/drive new product innovations, as described above under 'Streams'.

5) Drive data capabilities through machine learning, data mining and AI Adyen places a strong emphasis on machine learning, data mining and AI throughout the platform to improve the functionality and configuration of its core services. Aiming to maximize the authorization rates for merchants while minimizing the risk of fraudulent transactions, Adyen has developed a suite of data products to meet these needs (see "Products and Services – Data-enabled Products").

A key characteristic of the payments landscape is that it is dynamic – both fraudsters and industry players change their methods continuously and Adyen's data-driven approach allows it to react quickly to changes in the industry.

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Data Centres

WHY THIS MATTERS

Physical infrastructure and redundancy — the substrate under reliability scoring and failover. Pair it with the hop state machine and forward-correction model from Part III Ch2–3.

Risk mitigating measures Adyen has defined the following risk responses to manage the identified risks and holds own funds against the remaining residual risk. Where applicable, such measures comply with the relevant rules and regulations. See also the chapter "Supervision and Regulation".

Platform and system performance All of Adyen's services run on one secure platform, independent of whether payments are submitted online, mobile or through POS terminals. Adyen develops and maintains its flexible and entirely custom-designed platform and software in-house with a large development team.

Cyber security The Management Board has appointed a security officer. The security officer is responsible for maintaining and managing security within Adyen. Adyen's information security team performs an annual information security risk assessment to identify potential threats and control measures. A

formal and documented information security policy that includes guidelines on integrity, confidentiality and availability of information is available and implemented within the organization. Various preventive and detective controls have been implemented, including monitoring tools for the payment platform and databases to detect unusual activities. Furthermore, a cybersecurity insurance policy has been put in place. Adyen has separated the payment platform and customer area from Adyen's public website. Adyen currently operates six live data centers at different continents so that traffic can be rerouted real-time whenever needed and is planning to establish additional data centers. Exceptions in uptime connection, as well as the connection with other acquirers, are monitored continuously.

Money laundering and terrorism financing Adyen only accepts payment methods that are legal and generally accepted. New merchants are screened in accordance with Adyen's know your-customer ("KYC") procedures using a risk-based approach in which the risk of money laundering, terrorism financing activities and circumvention of sanctions is assessed. The identity of the merchant, ultimate beneficial owners and the nature of the business are verified with source documents and Adyen checks whether any of these may be a Political Exposed Person (PEP). Existing merchants, their legal representatives and ultimate beneficial owners are also regularly checked against sanctions lists. All transactions processed through the payment platform are monitored through automated rules to identify exceptional patterns, which may flag suspicious activity.

See also the chapter "Supervision and Regulation" under "Integrity".

Sanctions

New merchants are screened in accordance with the KYC procedures, using a risk-based approach in which the risk of circumvention of sanctions is assessed. The identity of the merchant, ultimate beneficial owners and nature of the business are verified with source documents and reviewed with sanctions lists (e.g. through World-Check). By using an automated routine (every 14 days), a review of all merchants, legal representatives and ultimate beneficial owners against sanctions lists is performed.

See also the chapter "Supervision and Regulation" under "Sanction Regulation".

Compliance with (foreign) laws and regulations Adyen has established global legal, compliance and regulatory affairs teams, which monitor compliance with (global) laws and regulations relevant to Adyen. The Management Board requests legal opinions from well-established law firms on local laws and regulatory requirements prior to starting foreign operations.

In addition, when setting up foreign operations, local management with knowledge of the local market is recruited. The legal, compliance and regulatory affairs teams strive to keep up to date with the latest regulatory developments and changes in laws and regulations are regularly analyzed and translated into updates of policies, procedures and systems.

See also the chapter "Supervision and Regulation" for a description of most important supervisory laws and regulations to which Adyen is subject.

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Our Global Network

WHY THIS MATTERS

A corridor network at scale — 1,700 corridors, localization, and a self-reinforcing data flywheel. This is the commercial analogue of corridor engineering and the compounding-data thesis running through Parts I–II.

Conveniently Putting Money Safely in the Hands of Our Customers' Families, Wherever They Are, by Relying on Our

Global Network

Our global network of funding and disbursement partnerships enables us to complete money transfers in over 1,700 corridors without the need to deploy local operations in each country. We are able to do this while complying with global and local licensing and regulatory requirements.

We have relationships with more than 15 top tier banks including Barclays, Chase, HSBC, and Wells Fargo, and leading global payment providers including a direct relationship with Visa. These relationships provide our customers an array of payment (or pay-in) options to fund remittances with a bank account, card-based payment, or alternative payment method. Our disbursement network provides our customers with a choice of various digital and traditional delivery methods and enables us to disburse (or pay-out) funds within minutes, or even seconds, to more than 3.5 billion bank accounts, over 630 million mobile wallets and alternative payment methods, and over 355,000 cash pickup locations (including retail outlets and banks). These partner relationships help drive a better customer experience, including faster transfers, higher acceptance rates, and enhanced reliability.

The map below illustrates the breadth of our global presence. Today, our customers primarily send money from the United States, Canada, the United Kingdom, other countries in Europe, and Australia, and our largest send country by revenue is the United States. Revenue from the United States represented \$199.0 million for the year ended December 31, 2020, which was 77% of total revenue for the year. On the other end of our global network, our customers' recipients are located in over 115 countries around the world; our largest receive countries include India, the Philippines, and Mexico.

The quality and diversity of our global network result in the following key advantages:

- **Global reach:** Today, our network enables us to complete cross-border payments in over 1,700 corridors from 17 countries to over 115 countries, in

over 75 currencies.

- **Local expertise:** We designed our global network to leverage local relationships for access into some of the hardest-to-reach markets around the world and to serve the unique financial needs of diverse immigrant communities. We help our customers send money and their families receive money in the manner they prefer and to which they are individually or culturally accustomed.
- **Control over the transaction lifecycle:** Our ability to manage each transaction from pay-in to pay-out is made possible with our direct integrations to 100 partners around the globe. These direct integrations help drive faster availability of funds to our customers' families, with more than 75% of total transactions in 2020 completed in less than one hour. Additionally, we have tools to reduce transaction declines or exceptions which help enhance conversion and facilitate faster processing. We can also optimize transaction routing for cost and risk and compliance management.
- **Security and compliance:** Our artificial intelligence ("AI") and machine learning-driven fraud detection and risk management engine is a foundational element that underpins our global network. We apply KYC and anti-money laundering standards that are tailored to meet local requirements of the jurisdictions where we operate. In addition, we leverage our in-depth knowledge of the markets in which we operate to execute tactically while complying with local licensing, compliance, and regulatory requirements. In 2020, we began serving business customers with the launch of Remitly For Developers, our remittance-as-a-service offering that strategically leverages our custom-built global network and compliance and regulatory infrastructure. With Remitly for Developers, businesses and their developers can integrate this network and infrastructure into their existing applications and websites through our Application Programming Interface ("API"). This enables them to offer digital cross-border remittances to their customers and introduce new digital banking solutions in emerging markets. Creating Trusted and Personalized Experiences with Our Localization Expertise at Scale Localization can mean many things. To us, it means speaking with our customers in their preferred language, reaching them through the media

channels they frequent, and being culturally relevant throughout their journey. While our business is global, we recognize the importance of a culturally-relevant experience being delivered to our customers and their families in the over 135 countries we serve. Our early success can, in part, be attributed to our localized approach within our initial corridors. As we have grown to over 1,700 corridors, we have found the appropriate balance of localization and scale by combining our customer-centric culture, investments in technology platforms, and data-driven decision making. Our localization approach enables us to provide customers with a personalized experience that drives peace of mind. We strive to deliver marketing, product, and support experiences that connect with them in meaningful ways:

- **Localized marketing at scale.** We achieve localized marketing at scale through a blend of deep cultural insights, consistent branding, rigorous analytics, and sophisticated channel management. We do this efficiently through our proprietary marketing technology stack and multi-faceted targeting techniques. As a mobile-centric business, we have the ability to leverage native app capabilities such as language preference, geolocation, and communication preferences to tailor the customer experience and fine-tune programs like promotions, referral campaigns, and activation campaigns.
- **Localized remittance product at scale.** We strive to deliver an exceptional product experience for not just our customers (on the send side), but also for their families (on the receive side). For our customers, we localize the experience based on their preferred language, the receive country, and the specific payment options from the send country. Based on those factors, we also tailor promotional offers, delivery speeds, foreign exchange (“FX”) rates, and delivery method preferences. This means understanding local payment and pricing norms, as well as customer and recipient expectations, and incorporating them into our services and prices.
- **Localized customer support at scale.** We deliver localized, and effective support and risk management to our customers around the world. We have a rich self-help center in-app and on the web with solutions to most customer issues available in 14 languages. Approximately 50% of our customer service questions are handled via self-help and automation. We also serve customers

via direct messaging in various social media channels. In order to best localize our support experience, our service centers are located in certain key receive countries or countries with language capabilities that are relevant to our customers. Using Our Data-Driven Approach to Better Serve Our Customers and Provide More Value Leveraging data is at the core of how we grow our business, optimize our customer economics, and prioritize our investments. We possess a unique, rich data asset with over ten years of transaction data. We monitor metrics at each step of the customer journey, and use this data to constantly improve the end-to-end customer experience.

- Data-driven platform. We have built a data platform that fuels analytics and drives meaningful customer insights. This platform enables us to aggregate data from multiple sources, including customer interactions on our app and the entire transaction processing life cycle. In addition to a robust data platform, we have a “build-measure-iterate” mentality that further aims to optimize our customer experience.
- Data-driven approach to customer acquisition. Establishing sustainable and attractive customer economics fuels our customer acquisition strategy. We manage our Customer Acquisition Costs (“CAC”) to corridor-specific targets, which are based on customer lifetime value. We set our targets based on data that we collect in each corridor including market maturity and opportunity, market awareness of our brand, and the incremental CAC we are seeing in our marketing channels. As we improve customer lifetime value through product enhancements and changes to operating costs and pricing, we are able to invest more in marketing while maintaining our marketing efficiency.
- Data-driven approach to managing the customer experience. Removing friction and ensuring a smooth onboarding process are key pillars to acquiring new customers and ensuring they become loyal customers who refer others. We utilize machine learning and data science to identify areas of friction, provide fair and transparent corridor-specific pricing, and optimize our compliance process:
 - Risk management. Our sophisticated risk management system enables us to identify and cancel fraudulent transactions efficiently, often in real-time. This allows us to mitigate our exposure to fraud loss while providing a low-friction

customer experience and minimize the sidelining of our good customers. We also constantly evaluate and integrate new risk management tools to provide our immigrant customers with onboarding options that are tailored to their unique circumstances.

- Pricing. We strive to set prices that deliver a great value to every single customer, while simultaneously accurately estimating our customer long term value (“LTV”). To achieve this, we incorporate corridor-specific customer behavior, competitor data, and market dynamics to manage prices through a proprietary pricing engine.
- Treasury. Our pricing is closely aligned with our robust treasury program, which minimizes our trading costs and mitigates currency risk, while ensuring that funds are delivered on time. The treasury program leverages a proprietary platform that incorporates advanced currency-level forecasting algorithms to estimate future demand and optimize our trading.
- Customer support. While maintaining a high-level of multilingual customer support, we measure and set reduction goals for support contacts per transaction, and we analyze that data in detail to prioritize improvements to our policies and services.
- Data-driven approach to customer loyalty. We analyze our customer’s behavior from sign-up to first send to becoming a repeat customer. We manage customers by cohort including sign ups, active customers, inactive customers and lapsed customers to spot trends in behavior. We Benefit from a Powerful Flywheel As we grow our customer base and complete more transactions, we collect more data. This data enables us to refine our marketing strategy, improve the customer experience, and accelerate our pace of innovation including introducing new services to our customers, or the recipient. Having a broader suite of services attracts more customers and enhances the experience, which could drive more transactions to Remitly and fuel further compounding organic growth.

Ability to Manage Risk and Fraud

WHY THIS MATTERS

Proprietary ML risk models, early-warning systems and bespoke rules kept within “desired guardrails” — the operating reality behind Aiva’s risk and compliance scoring engines (Part II Ch2).

Ability to Invest in Our Technology Platform and Introduce New Products

We will continue to invest significant resources in our technology platform. These investments will allow us continue to iterate, adapt, and add features to our current products, improve the customer and recipient experience, grow our payment and disbursement network, enhance our risk and security infrastructure, and continue to secure data in accordance with changing best practices and legal requirements, as well as to introduce new and innovate products. We expect our expenses related to technology and development to increase, which may impact short-term profitability, but we believe these investments will ultimately contribute to our long term growth.

Ability to Manage Risk and Fraud

We manage fraud (e.g., through identity theft) and other illegitimate activity (e.g., money laundering) by utilizing our proprietary risk models built on machine learning processes, early warning systems, bespoke rules, and manual investigation processes. Our models and processes enable us to identify and address complex and evolving risks in these unwanted activities, while maintaining a differentiated customer experience. In addition, we integrate historical fraud loss data and other transaction data into our risk models which helps us identify emerging patterns and quantify fraud and regulatory and compliance risks across all aspects of our customer interactions. This allows us to achieve and maintain fraud loss rates within desired guardrails.

Macroeconomic and Geopolitical Changes

Global macroeconomic and geopolitical factors, including immigration, trade and regulatory policies, unemployment, foreign currency fluctuations, and the rate of digital remittance adoption impact demand for our services and the

options that we can offer. These factors evolve over time and periods of significant currency appreciation or depreciation, whether in send or receive currencies, changes to global migration patterns, and changes to digital adoption trends may shift the timing and volume of transactions using our service.

Impact of COVID-19

The COVID-19 pandemic has caused significant disruption worldwide and many of our customers and employees have been impacted. With travel restrictions and shelter-in-place policies, the demand for digital remittances has increased, and this has driven a significant acceleration in our new customer growth.

We have also experienced, and may continue to experience, a modest adverse impact on our business practices, including as a result of transitioning part of our workforce to work from home and establishing strict health and safety protocols for our offices. Our customer support and operations teams, both internal and third-party, have been impacted, which has affected our ability to service customer needs due to longer wait times and our ability to hire personnel quickly.

Certain operating expenses have grown more slowly due to reduced business travel and the virtualization or cancellation of events. While a reduction in some operating expenses may have an immediate positive impact on our operating results, we do not yet have visibility into the full impact this will have on our business longer term. As COVID-19 vaccination rates increase and people begin to return to offices and other workplaces and travel more, the positive impacts of the COVID-19 pandemic on our business may slow or decline.

The full extent to which the COVID-19 pandemic will directly or indirectly impact our business, results of operations, cash flows, and financial condition will depend on future developments that are highly uncertain and cannot be accurately predicted. We do not yet know the full extent of potential impacts on our business or operations.

We will continue to actively monitor the situation and may take further actions that may alter our business practices as may be required by federal, state, or

local authorities or that we determine are in the best interests of our employees, customers, or business partners.

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Our Technology Platform

WHY THIS MATTERS

A vertically-integrated single stack — pay-in to pay-out, treasury ML, a custom ledger system and a “perfect delivery promise.” It maps almost one-to-one onto Lupine’s execution pipeline, treasury logic and evidence chains across Parts III–IV.

customer behavior, competitor data, and market dynamics to manage prices through a proprietary pricing engine. Our pricing engine also utilizes machine learning algorithms to identify key pricing levers within each corridor, enabling nuanced prices at scale. This customer-first, tech-enabled, and data-driven approach to pricing allows us to quickly identify discrete customer use cases, the product features most valued by those customers, and then price them accordingly.

- **Treasury.** Our pricing is closely aligned with our robust treasury program, which minimizes our trading costs and mitigates currency risk, while ensuring that funds are delivered on time. The treasury program leverages a proprietary trading platform that incorporates advanced currency-level forecasting algorithms to estimate future demand and optimize our trading. The strong link between our treasury and pricing programs helps minimize the impact of currency volatility on our revenues and our customer experience.
- **Customer support.** While maintaining a high-level of multilingual customer support, we measure and set reduction goals for support contacts per transaction, and we analyze that data in detail to prioritize improvements to our policies and services. We also measure and continuously improve classic support metrics around quality and efficiency of digital (self-help, chatbot)

and human support. **Data-Driven Approach to Customer Loyalty** We analyze our customer's behavior from sign-up to first send to becoming a repeat customer. We manage customers by cohort including sign ups, active customers, inactive customers and lapsed customers to spot trends in behavior. For corridors with lower performing product metrics (e.g., like order completion rate or second time transaction rate), we prioritize investments that drive better customer experience, which increases customer lifetime value and enables us to increase our marketing investment. With actionable data informing improved engagement and retention, we see higher LTV, customer satisfaction, and organic growth rates that lead to higher marketing efficiency. **Our Technology Platform is at the Core of Everything We Do** Our technology platform powers our mobile-centric suite of products, enables us to localize our product and experiences, connects our global network, and drives our data-driven approach. Our technology platform, which includes a combination of internally developed and third-party solutions, was built to be scalable, extensible, and seamlessly integrated. Our technology unites critical front- and back-end functions into a single, vertically integrated stack. This allows us to provide a low-friction customer experience, efficiently operate the business, and maintain control over product development. We utilize a cloud-hosted infrastructure which gives us cost and scale advantages. Everything is built with flexibility and configurability in mind, enabling our solutions to be dynamic as we continue to grow and expand into new geographies. We employ a Services Oriented Architecture to support fully decentralized, autonomous and high velocity engineering teams. **Enables Localization at Scale.** From customer acquisition through the disbursement of funds, many corridors display unique behaviors to which we tailor our technology. For example:

- Our platform enables marketing strategies to be executed with high efficiency and quality. We can intelligently tailor over 3,900 mobile- optimized landing pages by location, language, and promotion. We also test new marketing strategies and roll them out with high velocity. We use a turnkey A/B experimentation system to test different packaging, pricing, merchandising and user experiences, which enables us to learn from customer behavior and iterate.

- We can dynamically respond to a rapidly changing environment. Through our sophisticated marketing technology stack, we can set up new experimental channels in less than one week and launch new languages in less than one month.
- Once onboarded, our customers interact with our mobile app and website which are built to flexibly overlay one of our 14 local languages through automated and scalable translations. Highly Integrated Across the Transaction Lifecycle. We deliver simplicity for customers through a robust, scalable and flexible transaction engine that seamlessly integrates various functions across the transaction lifecycle. Our highly integrated platform enables interoperability of our strategic partners' multiple technology systems, functions and services through the entire remittance transaction from pay-in to pay-out. Key capabilities include:
 - Funding. Acquires funding via batch, instant and push payment methods.
 - Pricing. Using revenue optimization machine learning models and analysis, presents the exchange rate, fees, and promotional merchandising across the transaction lifecycle, from upper funnel marketing pages through the send experience.
 - Transaction processing. Orchestrates all system processes and functions into a single end-to-end payment flow.
 - Disbursement. Distributes funds through a diverse set of disbursement methods including bank deposit, mobile wallet, cash pickup, and home delivery. This system supports numerous integration approaches from traditional data file exchange and batch processing, to real-time push and pull APIs, all with the goal of creating a consistent and seamless customer experience regardless of disbursement method or partner integration approach.
 - Perfect Delivery Promise. Calculates the date and time of when the funds will be available to the recipient, a complex calculation done in real-time and made available to both our customer and their recipient. We provide a money back guarantee on timely delivery, a feature we are incredibly proud to stand behind.

- **Enabling technology.** Tightly integrates business operations processes across the customer journey to yield the best customer experience. Custom-built CRM solutions have use case specific workflows that are integrated with product delivery for seamless support at any stage of the transaction. Our treasury decision support system, supported by machine learning models, enables treasury operators to fund over 100 markets at scale ensuring a customer's money is delivered reliably and on-time. Our custom-built ledger system enables fast and scalable reporting, including automated tracking and creation of ledger lines, losses, chargebacks / disputes, and financial-related transaction events. Data-Driven Approach. All of our technology systems collect and manage data. We leverage this data to continuously test and iterate to optimize the customer experience and drive our various machine learning models.
- **Marketing.** We synthesize third-party analytics with Remitly website customer behavior to support channel-level spend optimization.
- **Pricing and product.** We use both internal and third-party licensed data to build and optimize our machine learning-based pricing models at scale across thousands of corridors. Using our proprietary A/B experimentation system, we execute, test, learn, and iterate cycles to improve our product offering, merchandising and customer experience.
- **Fraud.** We use internal data to build and optimize fraud machine learning models to detect fraudulent and illegitimate transactions. These models work in tandem with advanced user experience techniques including third-party step-up verification technologies and internally developed in-product mitigations to reduce friction while enhancing outcomes, identification and exclusion of illegitimate activity and fraud control.
- **Treasury.** We manage our FX inventory and risk across the globe on a daily basis through proprietary treasury decision support software that uses machine learning models that forecast demand and an integrated account management system bringing together hundreds of currency accounts. Low-Friction Risk Mitigation.
- **Advanced risk management system.** The nature of our business subjects us to the ongoing risk of fraud and other illegitimate transactions, including,

money laundering, scams and transactions subject to sanctions. To address these risks, we have advanced risk management and identity verification systems that both efficiently manage KYC and other compliance obligations and identify potential fraudulent transactions. Because of the efficiency of these systems we are able to maintain compliance while delivering a low-friction customer experience with low transaction loss rates. Utilizing proprietary machine-learning algorithms and trusted third-party vendor software, our technology platform applies dynamic risk management measures by corridor that enable us to stay ahead of fraud and other types of unwanted activity. Additionally, we deliver a low-touch, risk-based KYC process including automated document upload and digital proof of account ownership. Our ongoing monitoring capability leverages multiple security layers, including third-party identity verification technologies and multi-factor authentication. We believe our systems are scalable to meet future growth and are flexible and agile, which enable us to promptly respond to emerging fraud scenarios and changes in regulatory requirements. For example, with respect to new fraud risks, early warning systems trigger alarms when anomalous behavior appears, allowing us to shut down fraud attempts before they have an opportunity to scale. Alternatively, in the case of changing regulatory requirements, our policy engine allows us to dynamically configure KYC requirements. We continue to invest in our risk management capabilities including 24x7 expert human oversight to monitor transaction traffic and to spot evolving risks.

- Local and global regulatory compliance. We designed our technology platform to operate efficiently in a highly complex and changing regulatory environment. We believe our technology and compliance expertise allows for a consistently low-friction customer experience, which new market entrants would likely find difficult to replicate. As we launch our services into new markets, we ensure that we have rigorously considered and tested compliance with local experts and regulators. In addition, we have high standards and apply a robust diligence process in selecting our global distribution partners, which allows us to leverage their local regulatory expertise. Acquiring the U.S. federal, state and international regulatory approvals necessary to operate as a money transmitter is a time-consuming and capital-intensive process. We

have developed disciplined controls and systems, powered by technology, to comply with the requirements of handling cross-border remittances. As a result, we are able to monitor money movement at every step of the process to both support our reporting obligations and keep our customers well informed of their transaction status, but also as a benefit to our customers. Technology and Development Investment. We invest substantial time, energy, and resources to ensure we have a deep understanding of our customers' needs, and we continually innovate to deliver value-added products and services through our platform. We therefore have significant resources dedicated to technology and development across multiple teams including product, engineering, customer research, design, analytics, compliance, marketing, and customer service. These teams are responsible for the design, development, and testing of our platform and services. We focus the majority of our investment on developing new functionality and further enhancing the usability, reliability, and performance of our platform and services. Our Market Opportunity We believe our addressable market is large, fragmented, underserved, and in the midst of being disrupted, while benefiting from powerful structural tailwinds. Based on estimates from the World Bank and the International Monetary Fund, the market for global money transfers was estimated to be approximately \$1.5 trillion in total migrant remittance inflow volume in 2020 (including both approximately \$702 billion in remittances in 2020 and \$850 billion through informal channels), and has been growing at an approximately 4% CAGR over the past decade, with digital volumes significantly outpacing the rest of the market. In our core serviceable available market, remittance flows to developing countries reached \$540 billion in 2020, according to the World Bank. To contextualize the size and importance of this market, according to the United Nations, global remittances are over two times larger than the amount of official development assistance ("ODA"). As of June 30, 2021, we captured approximately 1% of the total migrant remittance inflow volume in this fragmented market. In addition, we believe we have a tremendous opportunity to significantly expand our total addressable market by leveraging our trusted brand and offer a broader set of financial services to today's more than 280 million global immigrants and their families. Our Long-Term Growth Strategy Our strategy

is designed to invest in our key strengths and create new opportunities that generate even greater value for our customers. The key elements of our strategy include:

- Gain share in existing corridors.
 - Grow our customer base. We plan on expanding our marketing efforts across existing corridors to increase brand awareness with customers and highlight the value of our products and services. We believe this will attract new customers to try Remitly. As we grow our customer base, we expect to benefit from increased operating leverage in the business and more data and insight to enhance our models.
 - Increase customer engagement and drive repeat use. The majority of our customers use our products and services multiple times per month. To further strengthen our customer relationships and brand loyalty, we will continue to enhance our products and services and develop new features to tailor and personalize our customers' experiences. We will also continue to establish new disbursement partnerships and add new payment methods to enhance our cross-border payment remittance experience for current and potential customers in our existing corridors. We expect these initiatives will attract new customers and lead to larger and more frequent transactions across our growing customer base.
- Expand to new corridors and partner networks. While our global network spans across 1,700 corridors around the world, we have plans to increase our reach to thousands of additional corridors. We see an opportunity to generate value by expanding our remittance services, Passbook and Remitly For Developers more broadly in this way. We expect to leverage our data-driven approach to optimize our pricing, product features, marketing strategies, and customer economics as we expand and grow in these new geographies. Additionally, we expect to leverage our localization expertise and our technology platform to grow the number of disbursement, payment and other partners in our global network and increase the number of direct integrations with such partners.
- Continue expanding into broader financial services. We believe there is an enormous opportunity to create a more inclusive financial system that not only encompasses, but caters to the needs of immigrants. We believe that the

insights we gain about immigrant customers through our data-driven platform will enable us to play an important role in developing products and services that meet this opportunity. We will continue to invest in our platform, expanding our product and service offerings and our overall technological lead to actualize our opportunities in this area. For example, our Passbook app started with deposit services, but we plan to build new products and features within the app in order to provide solutions to a broader array of our customers' problems. We will also continue to leverage our global network and infrastructure. We believe that empowering businesses to build on top of our leading distribution and compliance infrastructure, such as through Remitly For Developers, attracts a new set of potential customers and expands economic opportunities in developing markets.

- Pursue strategic partnerships and acquisitions. While our main growth strategy has historically been organic, we may selectively pursue strategic partnerships and acquisitions to accelerate our growth objectives or to enhance our competitive position within existing and new products and markets. For example, in March 2021, we extended our partnership with Visa and integration of Visa Direct within our global network, providing our customers with real-time cross-border payments options to more countries around the world. Our Culture Our customer-focused vision gives us purpose and motivates us to consistently think bigger, act more boldly, and deliver exceptional services for our customers. Our ability to deliver on our vision begins with our culture and values. Visa Direct capability enabled through Remitly's financial institution partner. Actual fund availability depends on the receiving financial institution and region.